



**Verified Carbon  
Standard**

## UPENERGY - SOCIAL AND CLIMATE IMPACT PROGRAMME- UGANDA-1



Document Prepared by UpEnergy Group

|                      |                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b> | UpEnergy - Social and Climate Impact Programme- Uganda-1                                                                                                                                                                     |
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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

The project activity involves promotion and distribution of improved cooking stoves (ICS) in Uganda. The purpose of the project is to support clean cooking interventions that moves end-user to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking in Uganda.

Through this project, the distribution and installation of approximately 197,475 operational ICS will be undertaken for households in Uganda. The annual average emission reductions are estimated to be 522,492 tCO<sub>2</sub>e and over a 7-years period of are approximately 3,657,442 tCO<sub>2</sub>e.

In this grouped project, each ICS constitutes a single project activity instance which is in line with the definition stated in VCS Program Definitions version 4.4.

### **Location of the project**

The project activity is being implemented in Uganda. The details of the project location are provided in Section 1.12.

### **An explanation of how the project is expected to generate GHG emission reductions or removals**

The project activity is distributing improved cookstoves in the project area to support low-income households. The ICS substitutes the current common cooking on open fire. The ICS burns wood more efficiently thereby improving thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions

### **Scenario existing prior to the implementation of project activity:**

Prior to the implementation of the project activity the existing scenario recorded as the use of traditional cook stoves and the use of non-renewable biomass by the target population to meet similar thermal energy needs and which also cause indoor air pollution (IAP) in absence of project activity.

### **A brief description of the project activity implemented at the time of validation:**

The first project activity was implemented on 09-May-2022 under this grouped project activity. 21,683 project activity instances (ICS) as a part of project activity have been implemented till 31-August-2022.

### **Total emission reductions achieved in this monitoring period:**

During the Current Monitoring Period from 09-May-2022 to 31-August-2022 (First and last date included) the project activity has contributed 12,978 tCO<sub>2</sub>e GHG reductions.

| Audit Type                  | Period                           | Program                           | VVB Name                              | Number of years |
|-----------------------------|----------------------------------|-----------------------------------|---------------------------------------|-----------------|
| Validation/<br>Verification | 09-May-2022 to<br>31-August-2022 | Verified Carbon<br>Standard (VCS) | LGAI<br>Technological<br>Center, S.A. | 0.3 years       |
| <u>Total</u>                | 09-May-2022 to<br>31-August-2022 | Verified Carbon<br>Standard (VCS) | LGAI<br>Technological<br>Center, S.A. | 0.3 years       |

## 1.2 Sectoral Scope and Project Type

**Sectoral Scope:** 03 Energy Demand.

**Project Type:** Type II Energy Efficiency Improved cook stoves Projects

The project is a grouped project

## 1.3 Project Eligibility

The project involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program. As per VCS Standard Version 4.4, efficiency improvements in thermal applications (e.g., cook stoves) are not excluded, hence project is eligible under VCS.

Also as mentioned in the VCS standard Version 4.4 section 3.5.15, this project satisfies all the criteria's mentioned. The justification for all the criterions has been given in the section 1.4 project design "Eligibility Criteria".

## 1.4 Project Design

This is a grouped project

### Eligibility Criteria

Along with above points, the Inclusion of New Project Activity Instances follows below criteria Grouped projects provide for the inclusion of new project activity instances subsequent to the initial validation of the project.

| No. | Criterion                                                                           | How the current and new project activity instances will comply                                                       |
|-----|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1   | Meet the applicability conditions set out in the methodology applied to the project | All the current project activity instances and the new project instances added to this grouped project will meet the |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | applicability conditions set out in Section 3.2 of this VCS Joint PDMR where the target population of end users are residential /household and the ICS deployed is at least 25% of thermal efficiency.                                                                                                                                                                              |
| 2 | Use the technologies or measures specified in the project description.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The current project activity instances and the new project instances added to this grouped project will use efficient improved cookstoves (ICS) to be distributed in the project.                                                                                                                                                                                                   |
| 3 | Apply the technologies or measures in the same manner as specified in the project description.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | The current project activity instances and the new project added to this grouped project will use Improved Cookstove to be distributed in the project and it will replace traditional cookstoves in household. All the project technologies distributed under the project will have a unique serial number and ID in order to curb double counting of the stoves under the project. |
| 4 | Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The current project activity instances and new project instances added to this group will take place within Uganda only and subject to the same baseline scenario determined in Section 3.4.                                                                                                                                                                                        |
| 5 | Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | This current and also the new project activity instances added to this group will distribute the project technology at subsidized value and more details on additionality have been added in the section 3.5 below                                                                                                                                                                  |
| 6 | Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows:<br>1) Each project activity instance that exceeds one percent of the capacity limit shall be identified.<br>2) Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.<br>3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit. | Not Applicable                                                                                                                                                                                                                                                                                                                                                                      |

## 1.5 Project Proponent

|                   |                                         |
|-------------------|-----------------------------------------|
| Organization name | UpEnergy Group                          |
| Contact person    | Anantha Karthik Rajagopalan             |
| Title             | Director of Carbon Programme            |
| Address           | 19 Cybercity, 10th Floor, Raffles Tower |
| Telephone         | Ebene, Mauritius                        |
| Email             | T: +230-404-6000                        |

## 1.6 Other Entities Involved in the Project

|                     |                                                                |
|---------------------|----------------------------------------------------------------|
| Organization name   | UpEnergy Uganda Limited                                        |
| Role in the project | Project Implementer                                            |
| Contact person      | Lafelle Chu                                                    |
| Title               | Country Director                                               |
| Address             | Plot 3848, Rwakiseta Road (Off Kironde Road), Muyenga, Kampala |
| Telephone           | T: +256 393 516 685                                            |
| Email               | lafelle@upenergygroup.com                                      |

## 1.7 Ownership

The project ownership is with UpEnergy Group, as explained below:

A voluntary and irrevocable agreement signed by the end user (cookstove user) conferring project ownership to the project proponent shall be the evidence of project ownership.

Each ICS shall have a unique serial number linking it to the project activity and shall be necessary to establish the fact that the ICS belongs to this VCS programme. End users will be informed at the time of ICS installation/distribution that the ICS will be discounted on account of them generating carbon credits and also the fact that these credits belong to project proponent. The end user will be required to sign the carbon waiver form cum warranty card as required by the project proponent to surrender their rights to carbon credits generated by the project.

## 1.8 Project Start Date



09-May-2022 (The earliest date of commissioning date of a batch distributed for first project activity instance).

## 1.9 Project Crediting Period

Project crediting period type: **Renewable**

Start date of first crediting period: **09-May-2022**

End date of first crediting period: **08-May-2029**

Total number of years: **7 years, 00 months**

The project activity adopts renewable crediting period of 7 years which can be renewed for maximum 2 times.

Being renewable crediting period, total crediting period of grouped project activity is 21 years.

## 1.10 Project Scale and Estimated GHG Emission Reductions or Removals

| Project Scale |   |
|---------------|---|
| Project       |   |
| Large project | x |

| Year   | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|--------|--------------------------------------------------------------------|
| Year 1 | 272,315                                                            |
| Year 2 | 507,941                                                            |
| Year 3 | 703,983                                                            |
| Year 4 | 643,005                                                            |
| Year 5 | 578,817                                                            |
| Year 6 | 511,251                                                            |
| Year 7 | 440,129                                                            |

|                                 |           |
|---------------------------------|-----------|
| Total estimated ERs             | 3,657,442 |
| Total number of crediting years | 07 Years  |
| Average annual ERs              | 522,492   |

### 1.11 Description of the Project Activity

The purpose of the project is to support clean cooking interventions that moves end-user and reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking in Uganda.

Through this project, the distribution and installation of approximately 197,475 operational ICS will be undertaken for households in Uganda. It is intended that under this project residential users with Clean cooking technologies such as energy- efficient biomass Improved Cookstoves (ICS) will be distributed with the aim to reduce greenhouse gas (GHG) emissions from the burning of non-renewable woody biomass and/or charcoal for cooking. The ICS will burn fuel more efficiently thereby improving thermal transfer to pots, hence saving fuel. Not only will this reduce the rapidly progressing deforestation (In 2020, it lost 73.6 kha of natural forest, equivalent to 36.0 Mt of CO<sub>2</sub> of emissions)<sup>1</sup> in Uganda but will also reduce health hazards from indoor smoke pollution and women and children will have to spend less time collecting firewood. The end user will be informed in advance that the use of ICS generates carbon finance which in turn is used for subsidizing the price of ICS and for recovering project implementation costs.

More than 2.6 billion people in world lack access to clean cooking facilities, relying instead on solid biomass, kerosene or coal as their primary cooking fuel<sup>2</sup>. According to statistics, household air pollution, mostly from cooking smoke, is linked to around 2.5 million premature deaths annually worldwide<sup>3</sup>.

<sup>1</sup> <https://www.globalforestwatch.org/dashboards/country/UGA> (Last accessed 01/12/2022)

<sup>2</sup> IEA (2020), SDG7: Data and Projections Report, IEA, Paris <https://www.iea.org/reports/sdg7-data-and-projections> (Last accessed 01/12/2022)

<sup>3</sup> [https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health#:~:text=Each%20year%2C%203.2%20million%20people,air%20pollution%20data%20for%20details\).](https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health#:~:text=Each%20year%2C%203.2%20million%20people,air%20pollution%20data%20for%20details).) ((Last accessed 01/12/2022)

The lack of access to clean cooking remains very acute in sub-Saharan Africa with only 17% population having clean cooking access. The vast majority of people in sub-Saharan Africa thus rely on gathering biomass for cooking, in particular in rural areas, which dramatically damages health and impairs productivity improvements. Almost 490,000 premature deaths<sup>4</sup> per year are related to household air pollution from the lack of access to clean cooking facilities, with women and children the worst affected. Forest degradation, sometimes leading to deforestation, is another serious consequence of the unsustainable harvesting of fuelwood. Improved Cookstoves (ICS) improve heat transfer efficiency as compared to the baseline conventional stoves, thereby reducing both the amount of fuel used by ICS and GHG emissions from the burning of non-renewable woody biomass and/or charcoal.

This project will replace conventional inefficient stoves, with higher efficiency ICS to residential users by leveraging resources provided by the project activity. Therefore, in the absence of the project activity, the baseline scenario would be the traditional cooking technologies with use of non-renewable biomass/charcoal for meeting similar thermal energy needs share.

#### Technical Specifications of Smart Home Pro Cook Stoves for initial first project activity: -

|                                        |                   |
|----------------------------------------|-------------------|
| Model                                  | Smart Home Pro    |
| Type of Stove                          | Domestic          |
| Dimension                              | (ø 28 * 26 H) cms |
| Average Unit Weight, Kg                | 12                |
| Thermal Efficiency, %                  | 37.90             |
| Pot Type pot                           | Flat Bottom       |
| Fuel Type                              | Charcoal          |
| Estimated Life Span                    | 7 Years           |
| Average time to boil 5 litres of water | 33.8 minutes      |

The manufacturer may change for future project activity instances and the same will be transparently reflected during verification. The details and specifications of respective cook stoves models will be considered for ER calculation of project activity.

<sup>4</sup> IEA (2020), SDG7: Data and Projections, IEA, Paris <https://www.iea.org/reports/sdg7-data-and-projections> (Last accessed 01/12/2022)

The relevant implementation dates (e.g., dates of construction, commissioning, and continued operation periods).

|                             |             |
|-----------------------------|-------------|
| Installation of ICS         | 21,683      |
| Date of first ICS installed | 09-May-2022 |

**Project Developers/Representatives:**

|                                   |                         |
|-----------------------------------|-------------------------|
| Project Developer                 | UpEnergy Group          |
| Project Representative            | UpEnergy Group          |
| Additional Project Representative | UpEnergy Uganda Limited |

**Data collection of ICS end-user:**

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS is being assigned a unique serial number. This number will be recorded during the distribution process together with the following information (as appropriate and as available):

- Name and telephone number and address of recipient of the user
- Date of distribution
- Geographic area of sale
- Model/type of project technology distributed
- Quantity of project technologies distributed
- Unique identification alpha/numeric ID for each device that is distributed

The information collected is being stored in the electronic database excel sheet which is serving as project database for project monitoring and sampling purposes. The traditional cook stoves were operational and this was the scenario prior to the implementation of project.

## 1.12 Project Location

The project will be located within the boundary of Uganda having the geographical coordinates latitudes 4°N and 2°S and longitudes 29° and 35°E. It is part of Africa and the northern hemisphere.



Democratic Republic of Uganda

### 1.13 Conditions Prior to Project Initiation

Prior to the implementation of the project activity the existing scenario recorded as the use of traditional cookstoves and the use of non-renewable biomass by the target population to meet similar thermal energy needs and which also cause indoor air pollution (IAP) in absence of the project activity.

From 2001 to 2020, Uganda lost 918 Kha of tree cover, equivalent to a 12% decrease in tree cover since 2000, and 413 Mt of CO<sub>2</sub>e emissions which can also be accounted due to the use of traditional cookstove or inefficient cookstove which also increase the demand of fuelwood for burning.<sup>5</sup>

### 1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

<sup>5</sup> <https://www.globalforestwatch.org/dashboards/country/UGA> (Last accessed 01/12/2022)

There are no mandatory laws or regulations in the host country for the use of improved cookstoves in Uganda households. The project is a voluntary effort by the project proponent. A review is made on Uganda environmental laws and regulations as below:<sup>6</sup>

1. National Environment Action Plan (NEAP) and National Environment Management Policy (NEMP) of 1994.
2. National Forestry and Tree Planting Act (hereafter Forest Act), Laws of Uganda (Uganda Law Reform Commission)
3. The 1995 Constitution of Uganda
4. The National Environment Act, 1995

## 1.15 Participation under Other GHG Programs

### 1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program to avail carbon benefits during the crediting period.

### 1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

## 1.16 Other Forms of Credit

### 1.16.1 Emissions Trading Programs and Other Binding Limits

This project is not part of any emissions trading program or any other mechanism that includes GHG allowance trading.

### 1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit. The undertaking is submitted for the same.

*Has the project sought or received another form of GHG-related credit, including renewable energy certificates?*

☐ Yes

☒ No

---

<sup>6</sup> <http://www.lead-journal.org/content/07020.pdf> (Last accessed 01/12/2022)

### 1.16.3 Supply Chain (Scope 3) Emissions

Not Applicable

## 1.17 Sustainable Development Contributions

### 1.17.1 Sustainable Development Contributions Activity Description

**Project's contribution to Sustainable Development:** The contributions of proposed project activity towards sustainable development are explained with indicators viz. social, economic, environmental, technological well-being, legislative and temporal as follows:

**Environmental well-being:** The project activity will result in the reduction of fuel consumption and emission of greenhouse gases and thus conserve forest and biodiversity.

**Social well-being:** The project activity will pave the way for development and increases the social status and living conditions and the prevailing living standard in the vicinity of the project activity and thus results in empowering the nearby. Also, it will contribute to a small increase in the local employment by employing skilled and un-skilled personnel for operation and maintenance of the equipment. The project will reduce the drudgery of women, time saving and the use of saved time for other productive activities.

**Economic well-being:** The project has created a business opportunity during construction phase for local stakeholders such as suppliers, contractors etc. contributing to economic well-being aspects. Further, the project also influences creation of employment opportunities for local people, which would enhance their social status. Sufficiently enhance indoor air quality thereby improving health of women and children and reducing incidences of smoke and fire related injuries and therefore result in saving of health-related expenses.

**Technological well-being:** The proposed project activity will promote improved cook stoves that result in reduced fuel consumption and emissions due to cooking and heating water in homes.

**Legislative:** The Project Proponent has obtained all the relevant approvals required for the establishment and operation of the project activity.

### 1.17.2 Sustainable Development Contributions Activity Monitoring

| Row number | SDG Target | SDG Indicator                                                                             | Net Impact on SDG Indicator        | Current Project Contributions                                                                                                                                                                                                                                               | Contributions Over Project Lifetime                                                                                                                                                                                                                                              |
|------------|------------|-------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.         | 1.4        | Number of households having access to improved cooking technology due to project activity | Implemented activities to increase | The distribution of 21,683 energy efficient stoves under this project activity helped Uganda households by providing access to basic services                                                                                                                               | The project activity benefits 21,683 Uganda households by providing access to basic services.<br><br>Estimated number of ICS users over the entire project lifetime 1,184,850.                                                                                                   |
| 2.         | 7.1        | Number of households having access to improved cooking technology due to project activity | Implemented activities to increase | Number of ICS operational under the project activity are 20,478.                                                                                                                                                                                                            | During the current Project Activity 20,478 ICS are operational.<br><br>PP will ensure that the ICS are operational and all the ICS owners have the access to clean energy.<br><br>Number of estimated ICS operational over the entire lifetime of the project will be 1,125,607. |
| 4.         | 8.b.1      | Number of Trainings conducted in a year                                                   | Implemented activities to increase | The project activity provides training and skill development program for youth population, thus increasing their employability. A total of 1 training was held for the employees as a part of this monitoring period. This will increase in the upcoming monitoring period. | A total of 1 training was held for the employees for this project activity during monitoring period.                                                                                                                                                                             |
| 5.         | 8.5        | Number of jobs created                                                                    | Implemented activities to increase | This project activity has created 73 direct and indirect jobs for marketing / sales and distribution / technical employees during the current monitoring period.                                                                                                            | This project activity has created 73 direct and indirect jobs.                                                                                                                                                                                                                   |



|    |      |                                                       |                                    |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                            |
|----|------|-------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6. | 12.2 | Decrease in specific fuel consumption                 | Implemented activities decrease to | The distribution of ICS has helped to reduce the consumption of Non-renewable biomass in participant households to an extent 2.34 tonnes / year / HH                                                                                                                                       | The distribution of energy efficient stoves helped 21,683 households to save 2.34 tonnes of non-renewable biomass per year per household                                                                                                                                                   |
| 7. | 13.0 | Tonnes of greenhouse gas emissions avoided or removed | Implemented activities decrease to | <p>The project activity involves distribution of energy efficient cookstove. The ICS distributed will be more efficient, thereby improving thermal transfer to pots, hence saving fuel wood.</p> <p>The emission reduction from the current project activity is 12,978 tCO<sub>2</sub></p> | <p>The average annual GHG emission reduction over the lifetime of the project is estimated to be 522,492 tCO<sub>2</sub>e.</p> <p>During the current project activity emission reduction of 12,978 tCO<sub>2</sub>e from a period of 09-May-2022 to 31-August-2022 have been achieved.</p> |
| 8. | 15.1 | Amount of Non-renewable biomass saved                 | Implemented activities decrease to | The project activity has reduced 47,854 tonnes of non-renewable biomass over the course of the monitoring period in participant households which contributed towards reducing the rate of deforestation.                                                                                   | This project has saved 47,854 tonnes of non-renewable biomass and has contributed in reduction of deforestation activity in Uganda.                                                                                                                                                        |

## 1.18 Additional Information Relevant to the Project

### Leakage Management

There is no leakage for this project activity and the project activity is already including 0.95 as a discount factor.

### Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

### Further Information

The Project Proponent has obtained all the relevant approvals required for the establishment and operation of the project activity.

## 2 SAFEGUARDS

### 2.1 No Net Harm

No potential negative environmental or socio-economic impacts have been identified for the project.

The project activity aims to distribute improved cookstove in the project area which will result in reduced consumption of natural resources. Also, the project activity will lead to reduced smoke/PM emission in the households, increased employment and overall emission reduction in the project area. The project technology is distributed at subsidized price and also reduces economic pressure from buying of wood fuel. Thus, no negative socio-economic impact is expected out of this.

All the raw material used for the technology have been procured sustainably and no negative impact on the natural resources is expected from this sourcing.

### 2.2 Local Stakeholder Consultation

The procedures or methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was sought).

The Stakeholder Consultation meeting was held on the 18th September, 2021 with the participation of different stakeholder categories. There were about 19 participants from various categories participated in the LSC meeting such as:

- ✓ Local people, communities and/or representatives who are expected to be directly or indirectly affected (adversely affected or beneficiaries) by the project or may have an interest in the project
- ✓ Stakeholders with land-tenure rights within or adjacent to the project and marginalized individuals and groups
- ✓ Local policymakers and representatives of local authorities
- ✓ National government officials or National Focal Point, for example, Designated National Authority (DNA) or equivalent body
- ✓ Local non-governmental organizations (NGOs), Women Groups working on topics relevant to the project or working with communities who are likely to be affected by the project

The consultation process included invitation to wide range of invitees to include effective and equal participation of both men and women. But due to COVID, the attendees were less. Prior to consultation, UpEnergy had provided with the following documents in the language that allows local stakeholders to understand and engage with the project.

- ✓ Non-Technical summary with relevant information
- ✓ Summary of the economic, social and environmental impacts of the project
- ✓ Contact details of the UpEnergy group to get for getting technical and project related information of the project
- ✓ Email id of UpEnergy was provided to the stakeholders who are not able to join the consultation meeting

During the meeting, the Key Project Information was presented to the local stakeholders. The sustainable development indicators and the monitoring of the indicators were clearly explained, and the Blind Assessment and Consultation Feedback Forms were given to them. Additionally, a copy of the presentation was also shared with them in order to help them provide feedback to UpEnergy Group in adapting the design and implementation of project activity in order to maximize benefits for end-users.

Up Energy's aspirational products such as the Domestic Smart Home Stove were displayed and a demonstration on how to use them was done. It was also explained to the stakeholders that similar technologies may also be implemented by UpEnergy based on beneficiary's feedback.

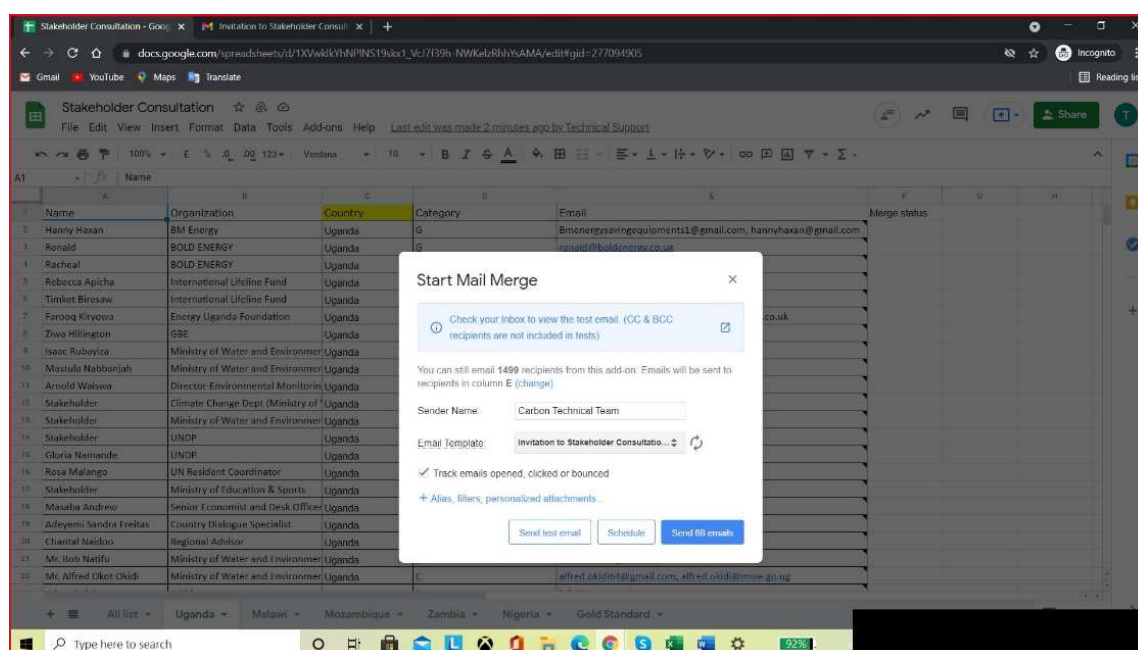
**The procedures or methods used for documenting the outcomes of the local stakeholder consultation.**

The stakeholder feedback round was initiated on 22<sup>nd</sup> September 2021. The reminder email for stakeholder feedback round closure was sent on 08<sup>th</sup> November 2021 as the last date for submitting the feedback was 22<sup>nd</sup> November 2021.

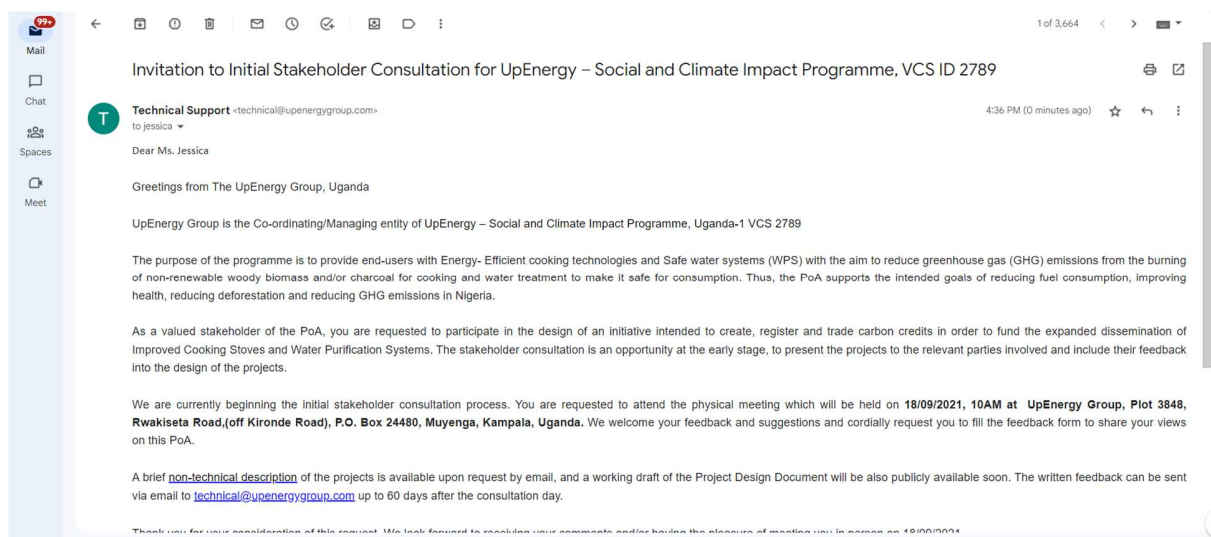
### Evidence proving invites took place as stated:

The invitation to stakeholders via email and screenshots of the same have been provided. In addition to email, the project proponent also reached out to stakeholders locally in Uganda to collect their feedback in person or telephonically.

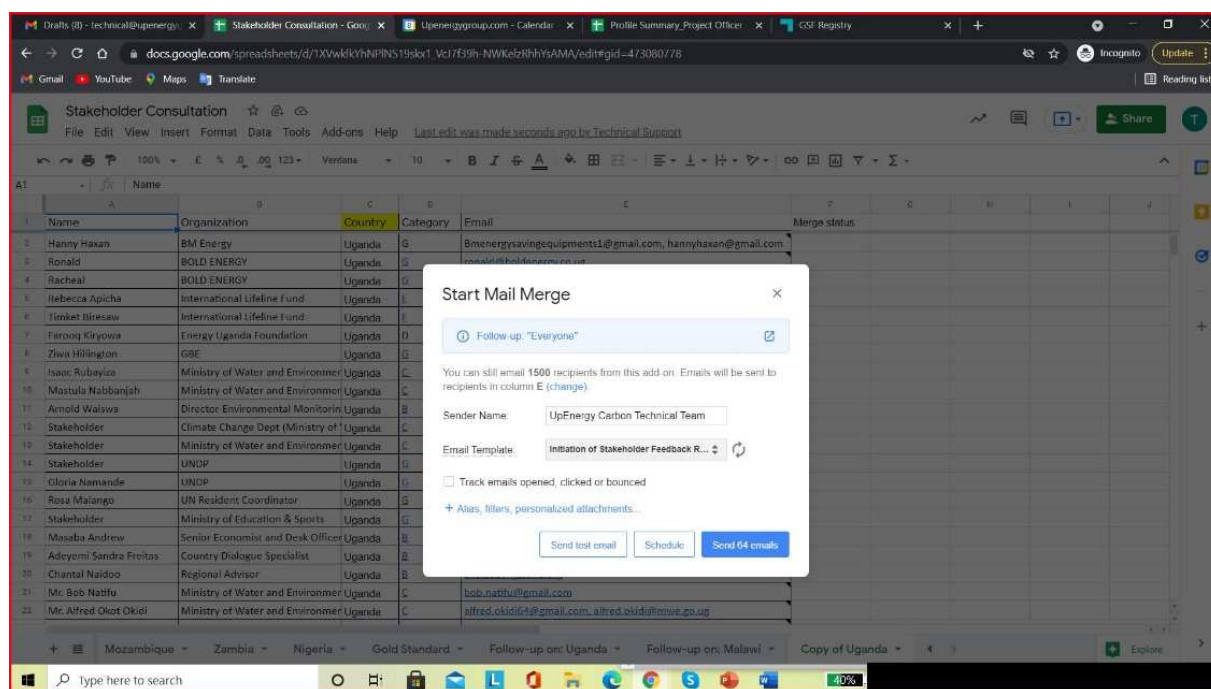
Screenshot of mail merge invite list sent to stakeholders on 19th August 2021



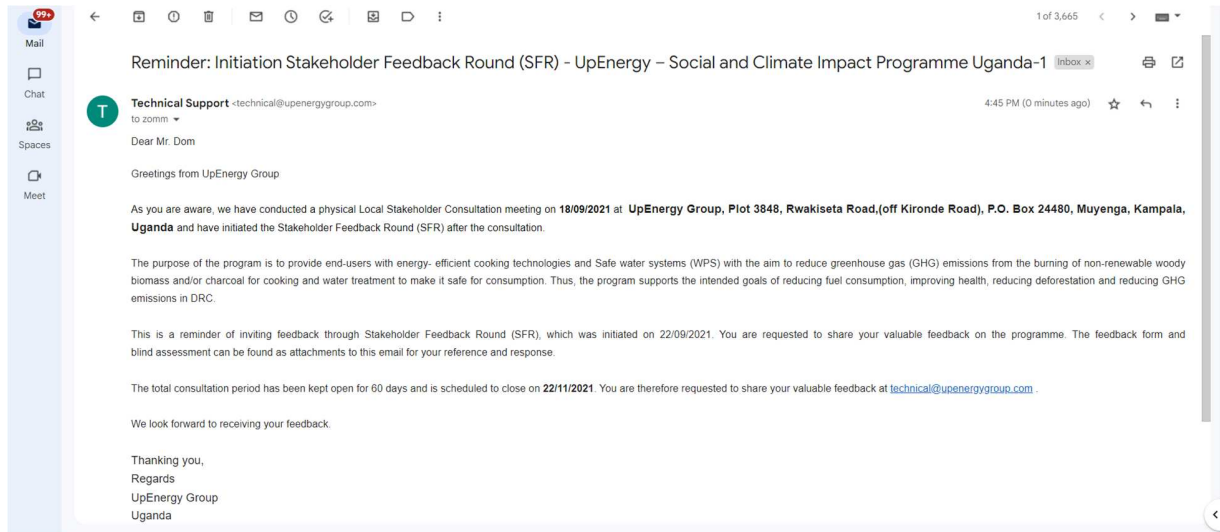
## Screenshot of 1<sup>st</sup> email sent to invitee



## Reminder Sent on 8<sup>th</sup> November 2021



## Screenshot of the remainder mail sent on 8<sup>th</sup> November 2021



## Newspaper publication:

Newspaper advertisement was given 2 times before the consultation meeting date. The same has been shown below



23<sup>rd</sup> August 2021



16<sup>th</sup> September 2021



## Minutes of consultations

The minutes of LSC has been recorded during LSC meeting which was held on 18<sup>th</sup> September 2021. There were 19 participants from various organization present in the LSC meeting. The meeting started by welcoming all participants followed by presentation from UpEnergy executive. The executive narrated the project's advantages, demonstration of ICS, SDG parameters, etc. Also, feedback forms and blind assessment forms have been given to the participants and collected the information. The meeting ended by 1:30 pm

The stakeholder feedback round had been initiated on 22<sup>nd</sup> September 2021.

Brief summary of the meeting:

- Meeting was called to order at 10am by meeting and chaired by Lead Data associate
- Associate started by welcoming all the stakeholders that managed to be present for the consultation, after members introduced themselves.
- Brief introduction on UpEnergy as a company, the carbon credits system, products that are on the market and company presence on the globe.
- Discussion on Smart home stove in depth and showed the stove parts and explained the different stove parts on the stove to the stakeholders.

In the introductory speech, the representative of UpEnergy Group, welcomed the gathering and given a brief about the climate mitigation project activity. Subsequent to the introductory speech, stakeholders were explained about the purpose of distribution of highly efficient cook stoves which will burn more efficiently thereby improving thermal transfer to pots, hence saving fuel. Project also reduce the rapidly progressing deforestation but will also reduce health hazards from indoor smoke pollution and women and children will have to spend less time collecting contributing to reduction in GHG emissions. The Project will also assist to the local population by providing employment opportunities to both skilled & unskilled labours.

## Assessment of comments from all consultations above:

| Gender of Stakeholder | Stakeholder comment                                                                   | Was comment taken into account (Yes/ No)? | Explanation (Why? How?)                                                    |
|-----------------------|---------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------|
| Male                  | Program should be rolled out to cover the entire country especially the area prone to | Yes                                       | The geographical area covered is all four administrative regions of Uganda |

|        |                                                                                         |     |                                                                                                       |
|--------|-----------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------|
|        | environmental degradation                                                               |     |                                                                                                       |
| Female | Carbon credits should be availed to all major green companies                           | Yes | The company is yet to decide on this.                                                                 |
| Female | Customers should be given caps, shirts, and money when visited for surveys              | Yes | We are providing shirt/caps etc for survey team                                                       |
| Male   | Should add more effort in the products/ publish in radio and televisions of the product | Yes | May consider in future after discussion with management                                               |
| Male   | Slow response                                                                           | Yes | Have the grievances mechanism and consider the point raised and try to reach out people alternatively |
| Male   | The moderator was knowledgeable                                                         | Yes | Narrated the technologies and operations etc to stakeholders in better way by live demonstrations.    |
| Female | UPE needs to understand properly the culture of our people and the challenges           | Yes | Already implemented this strategy and trying to improve as required.                                  |

The mechanism for on-going communication with local stakeholders:

|                                                                  | Method Chosen (include all known detail e.g. location of book, phone, number, identity of mediator)                                                                                                          | Justification of Choice (best practice)                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuous Input / Grievance Expression Process Book (mandatory) | Continuous input / Grievance Expression process book is available at the following office:<br><br>UpEnergy group<br>Plot 3848, Rwakiseta Road (off Kironde Road), P.O. Box 24480<br>Muyenga, Kampala, Uganda | Grievance Expression book will be placed at UpEnergy office in Kampala, Uganda. Stakeholders are free to voice their concerns via the Grievance Expression Book. By maintaining feedback book at the local office, it is ensured that stakeholders that don't have access to electronic media for expressing concerns / grievances are also able to share their concerns / feedback. Additionally, |



|                                           |                                                                              |                                                                                                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                                                              | the end users always have an option to revert to the salesperson (representative of distribution/retail partners etc.) in case of any feedback / complaints with the product post distribution. |
| Telephone access (optional)               |                                                                              |                                                                                                                                                                                                 |
| Internet/email access (optional)          | <a href="mailto:technical@upenergygroup.com">technical@upenergygroup.com</a> |                                                                                                                                                                                                 |
| Nominated Independent Mediator (optional) |                                                                              |                                                                                                                                                                                                 |
| Other                                     |                                                                              |                                                                                                                                                                                                 |

Overall, both stakeholders and beneficiaries unanimously acknowledged the benefits of the improved cookstove design and features and there was a considerable amount of positive feedback.

PP will ensure that regular monitoring is carried out by field staff and/or project managers in order to gather feedback from the local stakeholders. During the implementation of project activity, any stakeholders' concerns (both positive or negative) will be taken due care by the project proponent and necessary actions would be taken to resolve their queries.

There are no feedback comments received during the current monitoring period.

## 2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.<sup>7</sup>

## 2.4 Public Comments

No comments were received during the 30-day comment period which lasted from 08/06/2022 to 08/07/2022.

Link to the project: <https://registry.verra.org/app/projectDetail/VCS/2789>

## 2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

<sup>7</sup> <https://www.greenwatch.or.ug/greenwatch-publications/guide-environment-impact-assessment-process> (Last accessed 01/12/2022)

## 3 APPLICATION OF METHODOLOGY

### 3.1 Title and Reference of Methodology

Following approved baseline & monitoring methodology is applied.

Methodology: **VMR0006- Methodology for Installation of High Efficiency Firewood Cookstoves.**  
**Ver: 1.1.**

<https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

TOOL 01. Tool for the demonstration and assessment of additionality V07.0.0<sup>8</sup>

TOOL 30 Calculation of the fraction of non-renewable biomass EB 108 Annex 11 v4.0<sup>9</sup>

Standard - Sampling and surveys for CDM project activities and programmes of activities Version 9.0<sup>10</sup>

This methodology also refers the latest version of AMS II.G version 12 - Energy efficiency measures in thermal applications of non-renewable biomass.

<https://cdm.unfccc.int/methodologies/DB/GNFWB3Y6GM4WPXFRR2SXKS9XR908IO>

Sectoral Scope: 03.

### 3.2 Applicability of Methodology

| Applicability criterion                                                                                                                                                                                                                                                  | How the project complies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The project stove is a single pot or multi pot portable or an in-situ cookstove using only woody biomass; Additional requirement to demonstrate that the biomass used is solely renewable biomass in case project activities replacing baseline stoves using fossil fuel | <p>The proposed instances will distribute high efficiency single pot improved cook stove for thermal application, which will replace inefficient traditional cookstove leading to saving of non-renewable biomass. The ICS to be distributed under the project activity is Smart Home Pro. All ICS in this project will replace existing traditional stoves; therefore, no greenfield installations are included.</p> <p>Further, the project activity will utilize only renewable biomass under project scenario. This criterion will be checked from data recorded on the baseline stove used prior to ICS installation.</p> |

<sup>8</sup> <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

<sup>9</sup> <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

<sup>10</sup> [https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth\\_Stan05.pdf](https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf)

|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                   | Evidence – Baseline Scenario/Manufacture specification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Project activities shall be implemented in domestic premises or in community-based kitchen                                                                                                                                                                                                                        | The proposed instances involve distribution of ICS in residential/institutional/ commercial.<br><br>Evidence – End user agreement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| The project stove shall have specified high power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass. | Improved Cookstoves planned to be distributed under this project are SmartHome Pro stove that have an efficiency of 37.9% as per the manufacturer's specifications<br><br>The proposed instances do not involve replacement of fossil fuel-based stoves.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Both 'Projects' and 'Large Projects' can use this methodology                                                                                                                                                                                                                                                     | Estimated average annual emission reductions for the grouped project activity are greater than 300,000 tonnes CO <sub>2</sub> e per year. Therefore, proposed project qualifies the "Large Projects" criteria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;                                                                                                                                      | According to the U.N. FAO, 15.2% or about 2,988,000 ha of Uganda is forested, according to FAO (2020). Uganda had 51,000 ha of planted forest <sup>11</sup> .<br><br>Change in Forest Cover: From 2000 to 2020, Uganda experienced a net change of -1.05 Mha (-23%) in tree cover. From 2001 to 2021, Uganda lost 967 kha of tree cover, equivalent to a 12% decrease in tree cover since 2000, and 440 Mt of CO <sub>2</sub> e emissions. From 2001 to 2021, Uganda lost 967 kha of relative tree cover, equivalent to a 13% decrease since 2000 and 0.22% of the global total. <sup>12</sup><br><br>Annual Tree Cover Loss by Dominant Driver in Uganda: From 2001 to 2021, 3.9% of tree cover loss occurred in areas where the dominant drivers of loss resulted in deforestation. <sup>13</sup><br><br>Thus, it can be concluded that losses have been reported in forest areas from before 1989 thereby hinting that fuelwood used across Uganda comes from non-renewable sources. |
| For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood                                                                                                                                                                                                                             | Not applicable. The project stove which is improved energy efficient technology is introduced to replace baseline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

<sup>11</sup> <https://rainforests.mongabay.com/deforestation/2000/Uganda.html> (Last accessed 01/12/2022)

<sup>12</sup> <https://rb.gy/zbya7> (Last accessed 01/12/2022)

<sup>13</sup> <https://rb.gy/zbya7> (Last accessed 01/12/2022)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded. | stoves and reduce the use of non-renewable biomass for combustion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Additionally, applicability criteria numbers 8 and 9 set out in Section 2.2 of AMS II.G, version 11.1, later reflected under para 7 and 8 of AMS II.G, version 12 are demonstrated as below                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| The project design shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).                                                                                                                                                                                                                                 | Each ICS in this proposed instance will be identified by a unique combination of customer name and geographical location, as well as a serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project. In addition, the project has been cross-checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any other CDM project activity or voluntary project activity. |
| The project design shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices                                                                                                                                                                                                                   | ICS distributed under the project have a unique ID, and the ICS owners have transferred ownership of carbon credits to CME via end user agreement. Furthermore, that there are no other such projects in the region where the project intervenes, based on provided agreements with the project owner and distributors/producers and unique identification (serial number/logo) system on the ICS. As a result, it can be confirmed that the project boundary is clearly defined, and the technologies counted in the project are not included in another voluntary market or CDM project activity, avoiding double counting.                                                                                 |

#### TOOL01. Tool for the demonstration and assessment of additionality V07.0.0

| Applicability criterion                                                                                                                                                                                                                                                                           | How the project complies |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also | N/A                      |

|                                                                                                                                                 |                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| submit revisions to approved methodologies using the additionality tool.                                                                        |                                                                                          |
| Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory. | The tool is mentioned in VMR0006 in “step 3 project method” and is therefore applicable. |

### Tool 30 Calculation of the fraction of non-renewable biomass EB 108 Annex 11 v4.0

| Applicability criterion                                                                                                                                                                                                                                                                                               | How the project complies                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| This tool may be used by:<br>(a) DNAs to submit region-or country-specific default $f_{NRB}$ values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures);<br>or<br>(b) project participants to calculate project-or PoA-specific $f_{NRB}$ values. | As a project participant, the use of this tool is permitted. |

## 3.3 Project Boundary

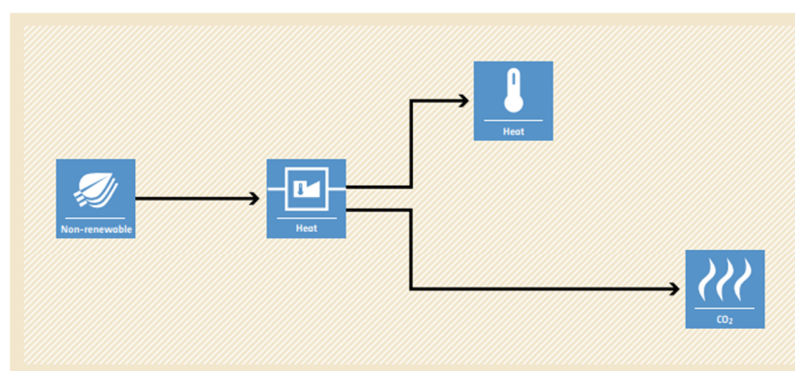
The project will be located within the boundary of Uganda having the geographical coordinates latitudes 4°N and 2°S and longitudes 29° and 35°E. It is part of Africa and the northern hemisphere.

| Source   | Gas              | Included? | Justification/Explanation  |
|----------|------------------|-----------|----------------------------|
| Baseline | CO <sub>2</sub>  | Yes       | Major Source               |
|          | CH <sub>4</sub>  | Yes       | Major Source               |
|          | N <sub>2</sub> O | Yes       | Major Source               |
|          | Other            | No        | No other source identified |
| Project  | CO <sub>2</sub>  | Yes       | Major Source               |
|          | CH <sub>4</sub>  | Yes       | Major Source               |
|          | N <sub>2</sub> O | Yes       | Major Source               |
|          | Other            | No        | No other source identified |

A representation of the baseline Scenario and project scenario are given as below Figure 3:

### BASELINE SCENARIO

Continuation of the current situation, i.e. use of nonrenewable biomass as fuel for the existing, less-efficient thermal applications.



**PROJECT SCENARIO** Installation of more-efficient thermal energy generation units utilizing non-renewable biomass and/or complete replacement of existing less-efficient thermal applications and/or retrofitting of existing thermal energy generating appliances reduces GHG emissions by saving nonrenewable biomass.

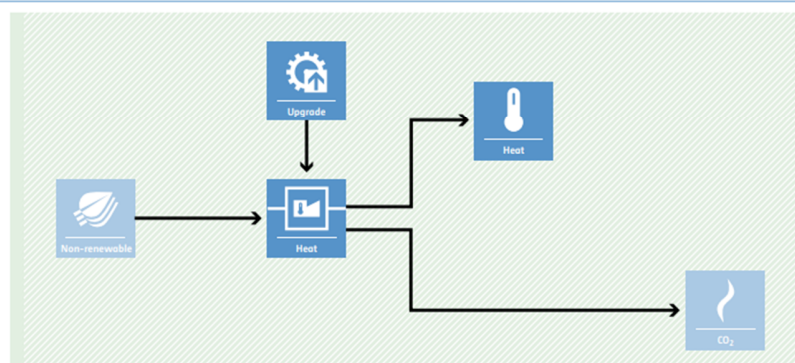


Figure 3: Baseline & Project scenario

## 3.4 Baseline Scenario

In line with the applied methodology, the baseline scenario is the continued use of non-renewable wood fuel (charcoal) and inefficient traditional cook stoves within the project boundary<sup>14</sup> by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.<sup>15</sup>

### Baseline Assessment by PP

UpEnergy Group has conducted a baseline assessment by interviewing 119 Households covering representative sampling at the national level by disaggregating the country into its four primary administrative regions: Central, Eastern, Northern and Western. These regions have become the natural boundaries for government statistical surveying and reporting regarding energy and fuel patterns. This also ensured that differences in geography would be captured in the survey.

<sup>14</sup> Charcoal and Wood Biomass Utilization in Uganda: The Socioeconomic and Environmental Dynamics and Implications, Sustainability 2020, MDPI

<sup>15</sup> Taking The Pulse of Energy Access In Uganda, Energizing Finance, 2019, page 105

Second, the survey employed a multistage random sampling approach to randomly choose districts within each region, and then randomly sampled counties within each district. Random sampling also ensured that the results captured the diversity of urban and rural communities which represent commonly observed fuel choices (ie. charcoal fuel more common in urban areas, and wood fuel in rural areas). However, at present the PP has developed local manufacturing capability for improved charcoal cookstove for this project where PP is intended to distribute charcoal-based ICS to replace charcoal-based traditional cookstoves. PP's intention is not to move traditional firewood stove users to improved charcoal stove users. The baseline fuel consumption (KPTs) were carried out targeting primary charcoal users using traditional stoves which is target baseline households planned for distribution.

The baseline scenario (traditional charcoal stove) can also be verified from the ICS distribution receipt at the time of distribution of the project device (ICS), which captures the baseline cooking technology & this can serve as a confirmation that, a registered user was using the traditional charcoal stove before adopting the project stove.

PP would like to state that, the current manufacturing of ICS is indigenous in nature, thereby encouraging the local economy & employment.

Based on the literature source, the potentiality has been identified for improvement in cooking technology and identified as base to target for baseline studies across Uganda.

### 3.5 Additionality

As defined by VMR0006, the activity method is used to demonstrate additionality.

#### **Step 1: Regulatory Surplus**

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by UpEnergy Group.

#### **Step 2: Positive list**

The applicability conditions of this methodology represent the positive list. The project must demonstrate that it meets all of the applicability conditions of the methodology as well as the below condition.

1. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.



2. Project activities that are implemented as part of government schemes or are supported by multilateral funds cannot be considered additional even if the stoves are distributed free of cost or at a highly subsidized rate and hence are not eligible to use this methodology

**Justification (1/2):** Since the project does not provide ICS installation to households free of charge and is not included in government schemes or supported by multilateral funds. PP has demonstrated additionality through step 3 project method as detailed below.

**Step 3:** Project method – Investment Analysis Tool -1 - “CDM tool for the demonstration and assessment of additionality” V07.0.0<sup>16</sup>

VMR0006 requires to conduct an investment analysis to “Determine whether the proposed project activity is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of verified emission reductions (VCUs).”

The PP selects option (b) to demonstrate that the proposed project activity is not economically or financially feasible without the revenue from the sale of verified emission reductions (VCUs).

#### 4.3 Step 2: Investment analysis

31. To conduct the investment analysis, use the following sub-steps :

##### 4.3.1 Sub-step 2a: Determine appropriate analysis method

32. Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis (Sub-step 2b). If the CDM project activity and the alternatives identified in Step 1 generate no financial or economic benefits other than CDM related income, then apply the simple cost analysis (Option I). Otherwise, use the investment comparison analysis (Option II) or the benchmark analysis (Option III).

Since the project activity for commercial approach generates revenues other than CDM related income, the PP can select either “option 2 Investment Comparison Analysis” or “option 3 Benchmark Analysis”.

##### 4.3.3. Sub-step 2b: Option II. Apply investment comparison analysis

35. Identify the financial indicator, such as IRR, NPV, cost benefit ratio, or unit cost of service (e.g. levelized cost of electricity production in \$/kWh or levelized cost of delivered heat in \$/GJ) most suitable for the project type and decision-making context.

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<sup>16</sup> <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>



The PP selects option 2, Investment Comparison Analysis with NPV as the financial indicator. The NPV is chosen since it is a direct assessment of the profitability and therefore a direct assessment of the financial and economic feasibility of the project. Since VMR0006 only requests the application of the Investment analysis (step 2), no identification and comparison with alternatives to the project activity is produced (step 1).

The NPV is calculated in the absence of revenues from carbon finance to demonstrate that the project is not financially and economically viable in those conditions ( $NPV \ll 0$ ). The NPV is calculated with the following main parameters: -

- ✓ An inflated discount rate of 6.8%<sup>17</sup> (for conservativity purpose).

The NPV is elaborated with the following income and expense data to establish a conservative analysis (all revenues included but not all costs):

- Income sources
  - Included: stoves sales
  - Excluded: carbon revenues
- Expenses sources:
  - Included: stove production/distribution/replacement, community services activities, selected part of direct operational costs (transport and warehousing, software etc....)
  - Excluded (for simplicity purpose, leading to a conservative NPV analysis): selected part of operational costs (salaries etc....), support costs: logistics, administration etc..., certification process costs.

The below tables summaries the key financial parameters and indicators followed by a detailed NPV<sup>18</sup> analysis for evaluation of project feasibility

Project Revenue and Costs: Over 7-year period.

| Project Cost                                  | Total in USD |
|-----------------------------------------------|--------------|
| Stove cost (7.42 USD/stove)                   | 1,567,156    |
| Implementation cost                           | 1,347,501    |
| Cost on account of defective stove/breakdown  | 111,047      |
| Costs involving software/database maintenance | 2,199,983    |

<sup>17</sup> <https://www.imf.org/external/datamapper/PCPIPCH@WEO/OEMDC/ADVEC/WEOWORLD/>

<sup>18</sup> Additionality has been worked out based on the cost manufacturing of lower cost ICS technology considering charcoal ICS and firewood ICS.

|                                                        |                   |
|--------------------------------------------------------|-------------------|
| O&M costs                                              | 5,223,088         |
| Total                                                  | 10,448,775        |
| Discounted cost                                        | 5,432,328         |
|                                                        |                   |
| <b>Project Revenue</b>                                 |                   |
| Revenue from sale of ICS                               | 795,822           |
| Other revenue                                          |                   |
| <b>Total</b>                                           | <b>795,822</b>    |
|                                                        |                   |
| Discounted annual revenue                              | 558,795           |
|                                                        |                   |
| <b>Net Present value excluding VCU's Revenue (USD)</b> | <b>-4,873,534</b> |

The NPV Sheet has also submitted to the VCS VVB along with this submission.

The NPV calculated is equal to -4,873,534 USD, clearly demonstrating economic unfeasibility without inflows from carbon revenues. An NPV analysis considering all expense sources excluded for simplicity purpose would show a project further unfeasible from a financial and economic point of view. The NPV sheet has been submitted along with this submission to the assessment team.

### Key Considerations

1. Project ICS are planned to be distributed at a subsidized price up to \$3.5 to promote clean cooking technology at Ugandan Households;
2. Discount rate is considered as 20% adopted in the company level
3. Inflation rate (6.8%) is sourced from the data provided by International Monetary Fund
4. Other operational costs / overheads are considered as per the actual company financial reports and also based on relevant management assumptions

A sensitivity analysis has also been carried out to understand and analyze the variations in the NPV with respect to the select key financial parameters viz., Cost of Manufacturing, ICS Distribution Cost and Inflation rate. These key financial parameters are selected for the sensitivity analysis considering the dynamics involved in market situation and economic conditions at the project location. The key outcome of the sensitivity analysis is tabulated below,

| Parameter / Variation | Project NPV in \$ |            |            |
|-----------------------|-------------------|------------|------------|
|                       | 10%               | 0%         | -10%       |
| Manufacturing Cost    | -4,987,714        | -4,873,534 | -4,759,353 |
| Inflation Rate        | -4,952,484        | -4,873,534 | -4,795,956 |
| ICS Distribution Cost | -4,817,654        | -4,873,534 | -4,929,413 |

As can be seen in the above table, the Project NPV is found to be negative in all iterations with respect to +/- 10% variations in the key financial parameters. This proves the project is additional in all scenarios and strongly indicates that the carbon revenue is vital for sustenance of the project.

#### Chronology of the Investment Decisions

| Event                          | Date       |
|--------------------------------|------------|
| Pre-Feasibility of the project | Dec 2021   |
| Investment Decision            | Jan 2022   |
| Date of the discount rate      | March 2022 |
| Start date of the project      | May 2022   |

#### Additionality conclusion

The investment analysis was conducted with conservative hypothesis and clearly showed impracticability from an economical and financial point of view. It is therefore concluded that the project activities are additional.

### 3.6 Methodology Deviations

The project activity has not taken any deviation in the applied approved methodology VMR0006 v1.1.

## 4 IMPLEMENTATION STATUS

### 4.1 Implementation Status of the Project Activity

During the current project activity, 21,683 ICS were distributed in the project area which includes 1<sup>st</sup> date of project technology sale from 09-May-2022 to end of 1<sup>st</sup> monitoring period date of 31-August-2022.

The model of stoves sold during this monitoring period were SHS Pro with a technical specificized efficiency of 37.9%. As a part of stove maintenance and after sales services, UpEnergy Uganda has a dedicated toll-free customer care number which can be used for any complaints or maintenance needed.

Some of the challenges faced during the implementation period were and their solutions provided by the project developer are listed below

1. Product defect due to either manufacturer error or HH misuse – For this the project developer has a robust warranty system with an operational customer care number
2. Poor data collection by field teams – Project developer always undertakes refresher training and also digitized data collection to avoid human error.
3. Supplier related shortages – have been overcome by onboarding various suppliers to ensure no reliance on only one source.

## 5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

### 5.1 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 5.4.

### 5.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 5.4.

### 5.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

## 5.4 Estimated Net GHG Emission Reductions and Removals

Net GHG emission reductions are calculated by applying Equations 1 and 2 for project activities replacing baseline stoves using non-renewable biomass (charcoal) and Equation 1 and 7 for project activities replacing baseline stoves using fossil fuel.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

**ER<sub>y</sub>**= Emission reductions during year y in t CO<sub>2</sub>e

**i** = Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire.

**j** = Indices for the situation where there is more than one batch of improved cook stove of type i

**ER<sub>y,i,j</sub>**= Emission reductions by improved cook stove of type i and batch j during year y in tCO<sub>2</sub>e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf,co2} + EF_{wf, non co2}) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where:

**B<sub>y, savings, i, j</sub>** = Quantity of woody biomass that is saved in tonnes per cookstove device of type i and batch j during year y

**f<sub>NRB, y</sub>** = Fraction of woody biomass that can be established as non-renewable biomass (f<sub>NRB</sub>)<sup>19</sup>  
(Calculation for FnrB done by using CDM tool 30)

**NCV<sub>wood fuel</sub>** = Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.0156 TJ/tonne)<sup>20</sup>

**EF<sub>wf, CO2</sub>** = CO<sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO<sub>2</sub>/TJ)<sup>21</sup>

**EF<sub>wf, non CO2</sub>** = non-CO<sub>2</sub> emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO<sub>2</sub>/TJ)<sup>22</sup>

**N<sub>y, i, j</sub>** = Number of project devices of type i and batch j operating during year y

**0.95** = Discount factor to account for leakage

<sup>19</sup> Default values endorsed by designated national authorities and approved by the Board are available at <https://cdm.unfccc.int/DNA/fNRB/index.html>

<sup>20</sup> AMS II.G. Version 12

<sup>21</sup> 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

<sup>22</sup> 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

The quantify of woody biomass saved  $B_{y, savings, i, j}$  due to implementation of improved cook stoves was estimated by following options set out in Equation 3.

$$B_{y, savings, i, j} = Bold \times \left(1 - \frac{\eta_{old}}{\eta_{new, i, j}}\right) \quad \text{Equation (3)}$$

Where:

**Bold** = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of Bold can be sourced from historical data or baseline surveys. A default wood to charcoal conversion factor of 6 kg of firewood per kg of charcoal may be used in line with paragraph 35 of AMS II.G, version 12. Alternatively, default value of 0.5 t/capita/year may be used.

$\eta_{old}$  = Efficiency of baseline cookstove

$\eta_{new, i, j}$  = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT). For this project activity efficiency is measured each monitoring period and annually.

Where the project households continue to use baseline cookstoves along with improved cookstoves, **Bold** shall be adjusted ex-post based on the percentage of project households found to continue such practice according to Equation 6. For such cases, the quantity of woody biomass saved  $B_{y, savings, i, j}$ , due to implementation of improved cook stoves shall be calculated using an adjusted value to account for ex-post use of baseline stoves in addition to improved cookstove

$$Bold_{adjusted} = Bold \times (1 - \mu_y)$$

Equation (6)

**Bold, adjusted** = Adjusted **Bold** to account the ex-post usage of charcoal in baseline cookstove(s) by project households in addition to improved cookstove (in tonnes per device).

$\mu_y$  = Baseline stove usage factor to account for use of baseline cookstoves along with improved cookstoves.

The quantity of charcoal consumed in absence of project activity (**Bold**) shall be determined using an estimation of average annual consumption of charcoal per household which may be derived using any of the following options:

- Historical Data:** Project proponent shall ensure that the relevance of data is appropriately justified for the target population and is the latest available data from credible source(s).
- Baseline Survey of Local Usage:** Project proponent shall carry out a survey of usage prior to implementation of the project activity following the sampling approach described in the latest version of CDM document Sampling and surveys for CDM project activities and programme of

activities. Alternatively, the project participant may follow the simple random sampling approach and the minimum sample size should be determined as per the following guidelines:

- Project target population < 300: Minimum sample size 30
- Project target population 300 – 1000: Minimum sample size 10% of group size
- Project target population > 1000: Minimum sample size 100

This simplified approach may also be used for determining minimum sample size for parameters listed under Sections 4.1 and 4.2 in which case it is not requisite for the sample size to meet confidence/precision requirements

- c. **Minimum Service Level.** Where historical data or a baseline survey has not been conducted, a default value of 0.5 ton/capita/year may be considered as the baseline biomass consumption. Household size shall be determined using credible references/literature or target population specific surveys. The survey shall be conducted as per guidelines outlined in option (b) above

In order to address the potential source of leakage which can be attributed to diversion of non-renewable biomass saved by project devices to non-project households which previously used renewable biomass; a net to gross adjustment factor of 0.95 is applied to  $ER_{y,i,j}$ .

#### Fraction of non-renewable biomass ( $f_{NRB}$ ) calculation for Uganda:

| Parameter                                  | Value       | Source                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Woody biomass consumption (t/yr) (H) | 133,545,800 | UBOS, 2019. Statistical Abstract, 2019. Uganda Bureau of Statistics, Uganda. <a href="https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf">https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf</a>                                                                                                              |
| Renewable biomass (t/yr) (RB)              | 14,583,019  | FAO, 2020. Global Forest Resources Assessment 2020 Report, Uganda. Food and Agriculture Organization of the United Nations. Rome, 2020. <a href="http://www.fao.org/3/cb0084en/cb0084en.pdf">http://www.fao.org/3/cb0084en/cb0084en.pdf</a><br>Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Local study (in accordance with section 3.2 para 19 of the TOOL30, v4.0) |
| Non-renewable biomass (t/yr) (NRB)         | 118,962,781 | Calculated (Equation 2 of CDM tool 30, v04.0)                                                                                                                                                                                                                                                                                                                                                           |
| Fraction of non-renewable biomass          | 0.89        | Calculated (Equation 1 of CDM Tool 30, v04.0)                                                                                                                                                                                                                                                                                                                                                           |

$$f_{NRB} = NRB / (NRB + RB)$$

Where:

fNRB = Fraction of non-renewable biomass in the applicable area in the relevant period (fraction or %)

NRB = Quantity of non-renewable biomass consumed in the applicable area in the relevant period (tonnes)

RB = Quantity of renewable biomass that is available on a sustainable basis in the applicable area in the relevant period (tonnes)

$$\text{NRB} = \text{H} - \text{RB}$$

Where,

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

$$\text{H} = \text{HW} * \text{N} + \text{CE} + \text{NE}$$

Where:

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes//household)

CE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE = Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

N = Number of households consuming wood fuel within the applicable area in the relevant period (number)

$$\text{RB} = \sum(\text{MAI}_{\text{forest},i} \times (\text{F}_{\text{forest},i} - \text{P}_{\text{forest},i})) + \sum(\text{MAI}_{\text{other},i} \times (\text{F}_{\text{other},i} - \text{P}_{\text{other},i}))$$

Where:

MAI<sub>forest,i</sub> = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas in the relevant period (tonnes/ha/yr)

MAI<sub>other,i</sub> = Mean Annual Increment of woody biomass growth per hectare in subcategory i of other land areas in the relevant period (tonnes/ha/yr)

F<sub>forest,i</sub> = Extent of forest in sub-category i in the relevant period (ha)

F<sub>other,i</sub> = Extent of other land in sub-category i in the relevant period (ha)

P<sub>forest,i</sub> = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in subcategory i) in the relevant period (ha)



$P_{other,i}$  = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category i) in the relevant period (ha)  
i = Sub-category i of forest areas and other land areas

Third party has used latest "Official statistics or reports or peer-reviewed literature" to derive the  $f_{NRB}$  value for Uganda (for detailed calculation refer the  $f_{NRB}$  report and analysis sheet).

Third party report also has been submitted to the VCS VVB.

#### Estimated Emission removal calculation for the parameters shown for project year 1 :

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$$

$$B_{old,adjusted} = 5.748 \times (1 - 0.1) = 5.17 \text{ tonnes wood/ year}$$

$$B_{y,saving,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}}\right)$$

$$B_{ySavings} = ((5.17) \times (1 - (18.32/33.75^{23}))) \\ = 2.37 \text{ tonnes wood/ year}$$

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf,co2} + EF_{wf, non co2}) \times N_{y,i,j} \times 0.95$$

$$ER_{y,i,j} = 2.37 \times 0.0156 \times 0.89 \times (112 + 26.23) \times 65,825 \times 0.95 = 301,604 \text{ tCO}_2\text{e}$$

Similarly for the upcoming years the calculation is shown in the table below:

| Year                                | Estimated baseline emissions or removals (tCO <sub>2</sub> e) | Estimated project emissions or removals (tCO <sub>2</sub> e) | Estimated leakage emissions (tCO <sub>2</sub> e) | Estimated net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|-------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|
| Year 1 (09-May-2022 to 08-May-2023) | 272,315                                                       | 0                                                            | 0                                                | 272,315                                                                |
| Year 2 (09-May-2023 to 08-May-2024) | 507,941                                                       | 0                                                            | 0                                                | 507,941                                                                |
| Year 3 (09-May-2024 to 08-May-2025) | 703,983                                                       | 0                                                            | 0                                                | 703,983                                                                |
| Year 4 (09-May-2025 to 08-May-2026) | 643,005                                                       | 0                                                            | 0                                                | 643,005                                                                |

<sup>23</sup> Efficiency of the device used in as the project technology calculated by using the formula given in VMR0006

|                                     |                  |          |          |                  |
|-------------------------------------|------------------|----------|----------|------------------|
| 08- May-2026)                       |                  |          |          |                  |
| Year 5 (09-05-2026 to 08-05-2027)   | 578,817          | 0        | 0        | 578,817          |
| Year 6 (09-May-2027 to 08-May-2028) | 511,251          | 0        | 0        | 511,251          |
| Year 7 (09-May-2028 to 08-May-2029) | 440,129          | 0        | 0        | 440,129          |
| <b>Total</b>                        | <b>3,657,442</b> | <b>0</b> | <b>0</b> | <b>3,657,442</b> |

The annual average of 7 years' period for project activity is 522,492 tCO<sub>2</sub>e

## 6 MONITORING

### 6.1 Data and Parameters Available at Validation

|                                                                                                     |                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data / Parameter</b>                                                                             | <i>Bold</i>                                                                                                                                                                                              |
| <b>Data unit</b>                                                                                    | Tonnes/year/HH                                                                                                                                                                                           |
| <b>Description</b>                                                                                  | Annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices. |
| <b>Source of data</b>                                                                               | Calculated through Baseline survey assessment & KPT                                                                                                                                                      |
| <b>Value applied:</b>                                                                               | Charcoal - 5.748                                                                                                                                                                                         |
| <b>Justification of choice of data or description of measurement methods and procedures applied</b> | This parameter is fixed ex-ante                                                                                                                                                                          |
| <b>Purpose of Data</b>                                                                              | Calculation of emission reduction                                                                                                                                                                        |
| <b>Comments</b>                                                                                     | Parameter Bold once determined shall remain fixed <i>ex-ante</i> for the entire crediting period.                                                                                                        |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Where charcoal is used by baseline devices, a default wood to charcoal conversion factor of 6 Kg of firewood per kg of charcoal may be used in line with paragraph 35 of AMS II.G, version 12.</p> <p>The project activity aims to replace the baseline traditional / inefficient charcoal stoves with the charcoal ICS, hence the baseline fuel consumption Bold = 5.748 tonnes/year/HH is applicable for project households that displaces charcoal</p> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                              |                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $f_{NRB,y}$                                                                                                                                                                                                                                              |
| Data unit                                                                                    | Fraction                                                                                                                                                                                                                                                 |
| Description                                                                                  | Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass                                                                                                                                   |
| Source of data                                                                               | Value determined from survey methods using the CDM tool 30 Version 4.0                                                                                                                                                                                   |
| Value applied                                                                                | 0.89                                                                                                                                                                                                                                                     |
| Justification of choice of data or description of measurement methods and procedures applied | This parameter shall be determined ex-ante. Default value as per Methodological tool: Calculation of the fraction of non-renewable biomass. As per of methodological tool, value is calculated as per following Equation (1), $f_{NRBy} = NRB/(NRB+DRB)$ |
| Purpose of Data                                                                              | Calculation of emission reductions                                                                                                                                                                                                                       |
| Comments                                                                                     | The report of $f_{NRB}$ will be made available to VVB during the validation.                                                                                                                                                                             |

|                                                                                              |                                                                                                                |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | $EF_{wf,CO2}$                                                                                                  |
| Data unit                                                                                    | tCO <sub>2</sub> /TJ                                                                                           |
| Description                                                                                  | CO2 emission factor for the use of wood fuel in baseline scenario                                              |
| Source of data                                                                               | 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion |
| Value applied                                                                                | 112                                                                                                            |
| Justification of choice of data or description of measurement methods and procedures applied | IPCC default value                                                                                             |
| Purpose of Data                                                                              | Calculation of emission reductions                                                                             |
| Comments                                                                                     | N/A                                                                                                            |

|                                                                                              |                                                                                                                |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | <b><i>EF<sub>wf,non CO2</sub></i></b>                                                                          |
| Data unit                                                                                    | tCO <sub>2</sub> /TJ                                                                                           |
| Description                                                                                  | Non-CO <sub>2</sub> emission factor for the use of wood fuel in baseline scenario                              |
| Source of data                                                                               | 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion |
| Value applied                                                                                | 26.23                                                                                                          |
| Justification of choice of data or description of measurement methods and procedures applied | IPCC default value                                                                                             |
| Purpose of Data                                                                              | Calculation of emission reductions                                                                             |
| Comments                                                                                     | -                                                                                                              |

|                                                                                              |                                                                                                       |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                                             | <b><i>NCV<sub>wood fuel</sub></i></b>                                                                 |
| Data unit                                                                                    | TJ/tonne                                                                                              |
| Description                                                                                  | Net calorific value of the non-renewable woody biomass that is substituted or reduced                 |
| Source of data                                                                               | 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction |
| Value applied                                                                                | 0.0156                                                                                                |
| Justification of choice of data or description of measurement methods and procedures applied | IPCC default value                                                                                    |
| Purpose of Data                                                                              | Calculation of emission reductions                                                                    |
| Comments                                                                                     | -                                                                                                     |

|                  |              |
|------------------|--------------|
| Data / Parameter | $\eta_{old}$ |
| Data unit        | Fraction     |

|                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                                                                  | Efficiency of baseline stove                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Source of data</b>                                                                               | Default value used from the methodology by combining it with weighted average of the two different type of baseline technologies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Value applied:</b>                                                                               | 20%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Justification of choice of data or description of measurement methods and procedures applied</b> | <p>As per the methodology where more than two types of baseline devices are replaced by the project technology weighted average values (taking the amount of woody biomass consumed by each device as the weighting factor) shall be used.</p> <ul style="list-style-type: none"> <li>- A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.</li> <li>- A default value of 0.2 shall be used for other types of devices.</li> </ul> |
| <b>Purpose of Data</b>                                                                              | Calculation of emission reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Comments</b>                                                                                     | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 6.2 Data and Parameters Monitored

|                                                                        |                                                                                                                                                                                                                                                                                                             |               |        |               |         |               |         |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------|---------------|---------|---------------|---------|
| <b>Data / Parameter</b>                                                | <b><math>N_{y,i,j}</math></b>                                                                                                                                                                                                                                                                               |               |        |               |         |               |         |
| <b>Data unit</b>                                                       | Number                                                                                                                                                                                                                                                                                                      |               |        |               |         |               |         |
| <b>Description</b>                                                     | Number of project devices of type i and batch j operating during year y                                                                                                                                                                                                                                     |               |        |               |         |               |         |
| <b>Source of data</b>                                                  | Monitoring Survey                                                                                                                                                                                                                                                                                           |               |        |               |         |               |         |
| <b>Description of measurement methods and procedures to be applied</b> | <p>Through Monitoring Survey</p> <p>Sampling standard will be used for determining the sample size to achieve 95/10 confidence precision according to the latest version of Standard for sampling and surveys for project activities and programme of activities.</p>                                       |               |        |               |         |               |         |
| <b>Frequency of monitoring/recording</b>                               | At least once every two years                                                                                                                                                                                                                                                                               |               |        |               |         |               |         |
| <b>Value applied</b>                                                   | <p>For ex-ante emission reduction calculation, it is assumed that the project will distribute ICS in the following scale:</p> <table border="1"> <tr> <td><b>Year 1</b></td><td>65,825</td></tr> <tr> <td><b>Year 2</b></td><td>131,650</td></tr> <tr> <td><b>Year 3</b></td><td>197,475</td></tr> </table> | <b>Year 1</b> | 65,825 | <b>Year 2</b> | 131,650 | <b>Year 3</b> | 197,475 |
| <b>Year 1</b>                                                          | 65,825                                                                                                                                                                                                                                                                                                      |               |        |               |         |               |         |
| <b>Year 2</b>                                                          | 131,650                                                                                                                                                                                                                                                                                                     |               |        |               |         |               |         |
| <b>Year 3</b>                                                          | 197,475                                                                                                                                                                                                                                                                                                     |               |        |               |         |               |         |

|                                |                                                                                                                                             |                |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                                | <b>Total</b>                                                                                                                                | <b>197,475</b> |
| Monitoring equipment           | N/A                                                                                                                                         |                |
| QA/QC procedures to be applied | -                                                                                                                                           |                |
| Purpose of data                | Calculation of emission reduction                                                                                                           |                |
| Calculation method             | Proportion of operational stoves ( $u_y$ ) obtained from the survey is multiplied by the total commissioned stoves to arrive at this value. |                |
| Comments                       | -                                                                                                                                           |                |

|                                                                 |                                                                                                                          |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter                                                | $\eta_{new, i, y}$                                                                                                       |
| Data unit                                                       | %                                                                                                                        |
| Description                                                     | Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y        |
| Source of data                                                  | WBT Test                                                                                                                 |
| Description of measurement methods and procedures to be applied | For the project activity efficiency of the device will be monitored by conducting water boiling test (WBT) annually.     |
| Frequency of monitoring/recording                               | Annually                                                                                                                 |
| Value applied                                                   | For ex-ante calculation, the value below is applied<br>SmartHome Pro – 37.9%                                             |
| Monitoring equipment                                            | -                                                                                                                        |
| QA/QC procedures to be applied                                  | Calibration details of the equipment mentioned in appendix 1. The WBT test has been conducted by third party laboratory. |
| Purpose of data                                                 | Calculation of emission reduction                                                                                        |
| Calculation method                                              | WBT Protocol v4.2.3 <sup>24</sup>                                                                                        |
| Comments                                                        | -                                                                                                                        |

|                  |         |
|------------------|---------|
| Data / Parameter | $\mu_y$ |
|------------------|---------|

<sup>24</sup> The Water Boiling Test Version 4.2.3 Cookstove Emissions and Efficiency in a Controlled Laboratory Setting

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data unit                                                       | Fraction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Description                                                     | Adjustment to account for any continued use of pre-project devices during the year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Source of data                                                  | Monitoring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description of measurement methods and procedures to be applied | <p>Minimum sample size of each type i and batch j will be in line with the guidelines provide in section 5.4</p> <p>This parameter will be monitored using the following method</p> <p>If both the improved cookstove and baseline cookstoves are used together then surveys will be conducted to record the average continued operation of baseline cookstoves in a sample of households. The surveys will be designed to capture the stove usage of households in the monitored region by formulating questions and/or collecting evidence to determine the frequency of usage of both the improved cookstoves and baseline cookstoves.</p> |
| Frequency of monitoring/recording                               | At least once every two years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Value applied                                                   | 0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Monitoring equipment                                            | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| QA/QC procedures to be applied                                  | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Purpose of data                                                 | Calculation of emission reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Calculation method                                              | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Comments                                                        | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### 6.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VMR 0006 version 1.1 on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

1. **User registers stove:** Project Proponent collected/received the necessary information required for the Registration process from the user. Means of collecting this information were through the use of ICTs. Project Implementers' staff double checks the accuracy of information provided, and request for field staff additional clarifications if needed;

2. **Data logged into database:** Registered data by the staff of the project proponent uploads in the database automatically. Project Proponent shall double check the information included on the database and check for duplications. Any duplicate information if investigated, checked for the errors, gets corrected or excluded from the database if it is a true duplicate entry.
3. **Spot- checking (ongoing):** Project Proponent field staff randomly select units included in the database and visit or contact the stove users to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) gets updated on the database, and in the case, ICS are found to be no longer in use, they got clearly marked as such and excluded from emission reductions calculations.
4. **Monitoring:** Project Proponent follows the requirements as per VCS-PD/MR to collect the necessary information for a monitoring report.

#### Baseline survey and baseline fuel consumption

UpEnergy Group conducted the baseline assessment by involving 119 randomly selected households from Uganda mainland including urban and rural households.

A total of 119 household were selected for the usages survey. The main sources of cook fuels used in most households are charcoal and a few other sources. The baseline study considered weightage of the samples corresponding to the population distribution in the different regions of the country (West 14%, North 17%, East 13%, Central 54%). Stratified representative sampling was applied at the national level by disaggregating the country into its four primary administrative regions: Central, Eastern, Northern and Western. These regions have become the natural boundaries for government statistical surveying and reporting regarding energy and fuel patterns. This also ensured that differences in geography would be captured in the survey.

The baseline KPTs were conducted for 45 households which use the charcoal as primary fuel across the above-mentioned regions. The No. of KPTs sample is in line with CDM Standard “Sampling and surveys for CDM project activities and programmes of activities” Version 9.0.<sup>25</sup>

| Fuel Type | Quantity of fuel consumed (MT/day/HH) | Quantity of eq. firewood consumed (MT/day/HH) |
|-----------|---------------------------------------|-----------------------------------------------|
| Charcoal  | 0.00262                               | 0.01575                                       |

#### Reliability Check for Charcoal KPT

|         |         |
|---------|---------|
| MEAN    | 0.0026  |
| MINIMUM | 0.00207 |

<sup>25</sup> [https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth\\_Stan05.pdf](https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf)



|                             |             |
|-----------------------------|-------------|
| <b>MAXIMUM</b>              | 0.0032      |
| <b>STANDARD DEVIATION</b>   | 0.00027     |
| <b>COVARIANCE</b>           | 0.10        |
| <b>Sample Size</b>          | 45          |
| <b>√n</b>                   | 6.71        |
| <b>Standard Error (SEy)</b> | 4.0907E-05  |
| <b>t-value (0.9, n-1)</b>   | 1.645       |
| <b>90% Confidence</b>       | 6.72861E-05 |
| <b>Precision attained</b>   | 3%          |
| <b>Compliance 90/10</b>     | YES         |

The average fuel consumption values established from the KPTs is **5.748 tonnes/HH/year**.

#### Sample Plan:

The monitoring plan is designed to monitor the parameters listed in the section above in Section 6.2. The share of operating stoves and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. UpEnergy Group will be responsible for conducting the sampling surveys and maintaining a database with all operating stoves.

No monitoring for leakage through competitive uses of biomass is required, as the parameter  $B_{old,i,j}$  is multiplied by 95% to account for leakage.

**As per the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04, the sampling plan is the following:**

#### (a) Sampling Design

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project activity that is designed in line with the requirements of the “Sampling and surveys for CDM project activities and programme of activities”, version 04.

#### (i) Objective and Reliability Requirements:

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period. As per Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities version 9.0, 95/10 confidence/precision shall be adopted for all parameters.

#### Monitored Parameters:

| Sr. no | Parameter          | Description of Parameter                                                             |
|--------|--------------------|--------------------------------------------------------------------------------------|
| 1      | $N_{y,i,j}$        | Number of project devices of type i and batch j operating during year y              |
| 2      | $\mu_{y,i,j}$      | Adjustment to account for any continued use of pre-project devices during the year y |
| 3      | $\eta_{new,y,i,j}$ | Efficiency of the device of each type i and batch j implemented as part of the       |

The efficiency of the project ICS ( $\eta_{new,i,j}$ ) is a monitoring parameter, which was calculated based on Water Boiling Tests (WBT).

## (ii) Target Populations:

The target population for the proportion of ICS still in operation ( $N_{y,i,j}$ ) in the project database which are using charcoal in ICS distributed under the project for cooking.

The target population for pre-project appliances ( $\mu_{y,i,j}$ ) is the set of old stoves still in use under project activity database.

## (iii) Sampling Frame

It is expected that the geographical locations do not have influence on the parameters of interest that are  $N_{y,i,j}$  and  $\mu_y$ . Therefore, these parameters can be assumed to be homogeneous regardless of how the end user group and distribution/installation location is defined since the end users shall always be households and geographical location is restricted to the project boundary. Since all stoves are of the same type, all stoves can be grouped into a homogeneous sampling frame.

### 1) Sampling frame for proportion of ICS in operation ( $N_{y,i,j}$ )

The sample frame refers to all the information sources on the Database. There are two primary mechanisms for data collection: the Registration Process for newly distributed/installed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the ICS. The detailed information collected from Registration Process is used to populate the stoves Database and the Monitoring Survey follows “Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 04.

As explained below (on section “Sampling Method”), to take the different characteristics of different project instances Implemented and ICS models into consideration, Project instances shall be grouped together to create a Primary Sampling Unit, which is homogenous. As per EB 86 Annex 04, Appendix-2, paragraph 1, for the use of a single sampling plan covering a group of projects, provided the homogeneity of population can be demonstrated, or differences are taken

into account in the sample size calculation, a 95/10 confidence/precision is applied for biennial sampling.

The first step is to identify the Primary Sampling Units. Primary sampling units are project instances, which have same ICS model. That is project with the same ICS model can therefore be grouped together and form a Primary Sampling Unit.

In case project has two different ICS models being implemented – this will form two Primary Sampling Units. This is justified by the fact that project might vary in terms of performance and it is important for the PP to collect and monitor accurate data for each stove model.

## **2) Adjustment to account for any continued use of pre-project devices during the year ( $\mu_{y,i,j}$ )**

During the same field surveys,  $\mu_y$ , the proportion of users still using their baseline stove, will also be determined. This method is preferred over using data loggers, since installing data loggers is not considered practical. Occasional usage of a pre-project stove is deemed insignificant, so  $\mu_y$  is only calculated if baseline stoves are used on a weekly basis.

This survey will be part of the above mentioned  $N_{y,i,j}$  survey and will therefore be conducted with the same households.

The surveys should capture the number of meals cooked using the baseline stove in the dry and wet season. There will be three situations:

1. Project ICS are completely discarded
2. Baseline stove(s) are completely discarded
3. Baseline stove(s) used along with project ICS.

The first scenario will not be taken into account for  $\mu$ , since this proportion of operational stoves is already used to calculate  $N_{y,i,j}$ . The second scenario gives a  $\mu$  of 0% for that household. For the third case, the number of meals cooked in the dry and wet season using the baseline stove will be divided by the number of meals cooked in the dry or wet season respectively. Then, a weighted average of the two will be taken to determine the baseline stove use percentage. The obtained percentages of all households in the second and third scenario will be averaged to obtain  $\mu_y$ .

## **3) Parameter $\eta_{new,i,j}$**

The thermal efficiency  $\eta_{\text{new},i,j}$  is a monitoring parameter and WBT testing has also been performed by third party (Centre for Community Research and Community Development Uganda) as below:

- (a) Conduct a sample test on cook stoves with 3 tests conducted for each stove.
- (b) The standard deviation of the test results indicated above is very small and 95/10 precision requirement is met. The efficiency determined is acceptable.
- (c) Efficiency tested is high-power thermal efficiency. The high-power thermal efficiency is the average of the Cold Start and Hot Start phases
- (d) The average of all results for each device type/model and batch shall be taken as the efficiency for each device type and batch.

A total of 33 project devices were selected based on random sampling. Further the WBT test were carried out from 13/10/2022 to 20/10/2022.

#### (iv) Sampling Method

The sampling method for monitored parameters  $N_{y,i,j}$  and  $\mu_{y,i,j}$  is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the Project Proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$ : Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$\mu_{y,i,j}$ : Pre project device only is in use then fraction to be used to calculate total number, however if pre project device is used along with project ICS, proportion of usage of each will be determined by cooking habits evaluated by survey questionnaire during the monitoring period.

Using the formulas in the section “Sample Size” below, the PP will **randomly sample** the required number of ICS from the primary sampling units. It is important to note that for  $\mu_{y,i,j}$  where partial usage of both old stoves and project ICS are observed, for each household under sample cooking habits must be taken into consideration.

#### (v) Sample Size

The estimation of the proportion or mean value of the parameters investigated, the minimum sample size for each sample frame has been calculated based on 95/10 confidence/precision which will suffice for biennial sampling.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 95 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from each Primary Sampling Unit.

There are two parameters for survey that are estimated through sampling: the number of stoves still in operation during the monitoring period as determined by the monitoring survey ( $N_{y,i,j}$ ), and the continued use of old stoves, ( $\mu_{y,i,j}$ ). In line with AMS-II. G, both can be sampled in a single survey with a random sample of households using the above-described confidence/precision levels depending on annual or biennial monitoring frequency.

Thus, both the sampling parameters considered based on 'proportion value', hence to estimate the sample size for parameters  $N_{y,i,j}$  and  $\mu_{y,i,j}$ . the following equation is used:

$$n \geq 1.962 N \times p (1-p) / (N -1) \times 0.1^2 p^2 + 1.96^2 \times p(1-p)$$

Where:

n = Sample size

N = Population size (Total number of households/ICS)

p = Expected proportion

1.96 = Represents the 95% confidence required

0.1 = Represents the 10% relative precision

### Minimum Sample Size

If the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 shall be chosen when the parameter of interest is a proportion.

### Quality Assurance/Quality Control

The UpEnergy Group will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system. The choice of measure applied to each parameter will depend on the cost of each data collection approach and logistics required. The UpEnergy Group will determine the most effective measure for each parameter: -

**Oversampling:** Randomly draw a sample more than the calculated number (say 20%) and collect data from each.

**Buffer Group:** Randomly draw additional samples (say 20%) and collect data from only for minimum numbers of ICS as per sample size calculation. If this would not result in the required sample size data would be collected from the additional ICS that were selected in the sample.

### Data archiving

Copies of the surveys will be kept and the registration database will have back up. Original stove purchase contracts or other means of acceptance by the users will be stored in the main office for the Project Proponent. A back-up of the registration database will also be stored on an electronic medium by the PP. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of VCU for the project activity, whichever is later.

### Personal Training

Separate training will be given to deal adequately with surveying techniques and reduce errors. The training includes implementing Water Boiling Test (WBT), efficiency of cookstove, frequency of survey that needs to be done.

### Roles and Responsibilities of Project Proponent

- ✓ Development of VCS project Description using latest template.
- ✓ Registration of the project with VERRA.
- ✓ PP itself shall officially communicate with VERRA for project registration.
- ✓ Providing improved cookstove (ICS) to replace traditional cookstove.
- ✓ Getting the project validated/verified by a Validation and Verification Body (VVB)
- ✓ Carrying out annual surveys and monitoring
- ✓ Maintaining the project database and other documentation required by VERRA
- ✓ Preparing the Monitoring Reports for the Project Activity

### Organizational Structure

| Person                    | Role                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PP database administrator | The database administrator is responsible for updating and maintaining all electronic databases. Required competencies include experience with data management systems (e.g. Excel, STATA, or ONA/ODK), minimum 2 years working experience in a similar field, and at minimum a Bachelor's degree from an institution of higher education. |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Monitoring team | <p>The monitoring team will be assigned by the PP to conduct the user interviews and check the operational status of Cookstoves (ICS) in Households during the periodic sampling and reports the results to the database administrator. The skills and experience required for the data collection activities include:</p> <ul style="list-style-type: none"> <li>• Experience conducting surveys/tests</li> <li>• Experience conducting door-to-door surveys</li> <li>• Local language skills (especially important for input to questionnaire design and interviewing of end users)</li> <li>• English language skills</li> <li>• Cultural awareness</li> <li>• Numerical proficiency</li> <li>• Data entry skills</li> </ul> |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Internal audits will be performed on a regular basis to ensure the consistency of monitored data. Prior to any verification, the PP will conduct a final check of the data and analyses project performance. All personnel involved in the Project Activity will receive the desired training for field measurements. The Project Developer will keep records of trained personnel. The Carbon Credit Manager will ensure that only trained personnel shall carry out the monitoring activity.

Any non-conformities identified during the internal audit; a non-conformity report is prepared which consists of the root cause of the non-conformity. Once the non-conformity is reported, Carbon Credit Manager in consultation with Head, Personnel prepares the corrective action measures and implements the same. After the implementation of the corrective actions the Internal Audit team conducts assessment on the non-conformity and if found ok, the Internal Audit closes the non-conformity.

## 7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

### 7.1 Data and Parameters Monitored

|                  |                                                                         |
|------------------|-------------------------------------------------------------------------|
| Data / Parameter | $N_{y,i,j}$                                                             |
| Data unit        | Number                                                                  |
| Description      | Number of project devices of type i and batch j operating during year y |
| Value applied:   | 20,478 (Total operational stoves)                                       |

|          |                                   |
|----------|-----------------------------------|
| Comments | Calculation of emission reduction |
|----------|-----------------------------------|

|                  |                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | $\eta_{new, i, y}$                                                                                                |
| Data unit        | %                                                                                                                 |
| Description      | Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y |
| Value applied:   | SmartHome Pro – 35.38%                                                                                            |
| Comments         | Calculation of emission reduction                                                                                 |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data / Parameter | $\mu_y$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Data unit        | Fraction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Description      | Adjustment to account for any continued use of pre-project devices during the year y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Value applied:   | 0.0647                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Comments         | <p>Calculation of emission reduction</p> <p>The usage survey for MP 1 has revealed out of 108 project ICS users 25 HHs have reported parallel usage of baseline stoves, thus implies 23.14% of the representative population uses both project and baseline technology.</p> <p>The survey also reveals that these 25 HH uses the baseline stove to an extent - meaning out of total cooking events 74% are carried out by project device and rest only 26% is being met by baseline stoves in these 25 HH.</p> <p><math>\mu_y</math> value represents this 26% dependency on the baseline stove, while extrapolating it to the entire surveyed population the weighted average baseline stoves usage factor is calculated to be 6.47% as shown below,</p> <p>Average baseline usage rate</p> <p>= (% of HH parallelly using baseline stove * Percentage of meals cooked on the baseline stove)</p> <p>= (23.14% x 26%) = 6.47%</p> |



## 7.2 Baseline Emissions

The methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 7.5.

## 7.3 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 7.5.

## 7.4 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology.

## 7.5 Net GHG Emission Reductions and Removals

Net GHG emission reductions are calculated by applying Equations 1 and 2 for project activities replacing baseline stoves using non-renewable biomass (charcoal) and Equation 1 and 7 for project activities replacing baseline stoves using fossil fuel.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

- **ER<sub>y</sub>**= Emission reductions during year y in t CO<sub>2</sub>e
- **i** = Indices for the situation where more than one type/model of improved cook stove is introduced to replace three-stone fire.
- **j** = Indices for the situation where there is more than one batch of improved cook stove of type i
- **ER<sub>y,i,j</sub>**= Emission reductions by improved cook stove of type i and batch j during year y in tCO<sub>2</sub>e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf,co2} + EF_{wf, non co2}) \times N_{y,i,j} \times 0.95$$

Equation (2)

Where:

- **B<sub>y, savings, i, j</sub>** = Quantity of woody biomass that is saved in tonnes per cookstove device of type i and batch j during year y
- **f<sub>NRB,y</sub>** = Fraction of woody biomass that can be established as non-renewable biomass (fNRB)<sup>26</sup>

<sup>26</sup> <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

- $NCV_{wood\ fuel}$  = Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.0156 TJ/tonne)<sup>27</sup>
- $EF_{wf, CO2}$  = CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO2/TJ)<sup>28</sup>
- $EF_{wf, non\ CO2}$  = non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO2/TJ)<sup>29</sup>
- $N_{y, i, j}$  = Number of project devices of type i and batch j operating during year y
- 0.95 = Discount factor to account for leakage

The quantify of woody biomass saved  $By_{savings, i, j}$  due to implementation of improved cook stoves was estimated by the following option set out in Equations 3.

$$By_{savings, i, j} = Bold \times \left(1 - \frac{\eta_{old}}{\eta_{new, i, j}}\right) \quad \text{Equation (3)}$$

Where:

- $B_{old}$  = Annual quantity of woody biomass that would have been used in the absence of the project activity (in tonnes per device) to generate useful thermal energy equivalent to that provided by the improved cook stove. The value of Bold can be sourced from historical data or baseline surveys. A default wood to charcoal conversion factor of 6 kg of firewood per kg of charcoal may be used in line with paragraph 35 of AMS II.G, version 12. Alternatively, default value of 0.5 t/capita/year may be used. For the current project activity Bold value has been derived from the baseline study conducted in the project boundary which is 5.807 tons wood-eq/HH-yr
- $\eta_{old}$  = As per the methodology where more than two types of baseline devices are replaced by the project technology weighted average values (taking the amount of woody biomass consumed by each device as the weighting factor) shall be used.
  - A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
  - A default value of 0.2 shall be used for other types of devices.

For the project activity the values used for baseline efficiency have been taken from the baseline study conducted on field in coordination with ICEED.

- $\eta_{new, i, y}$  = Efficiency of the improved cook stove type i and batch j determined through water boiling test (WBT). (Value applied for the current project activity is 35.38%)

<sup>27</sup> AMS II.G. Version 12

<sup>28</sup> 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

<sup>29</sup> 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

The quantity of fuel consumed in absence of project activity ( $B_{old}$ ) shall be determined using an estimation of average annual consumption of firewood per household which may be derived using any of the following options:

Baseline Survey of Local Usage: Project proponent shall carry out a survey of usage prior to implementation of the project activity following the sampling approach described in the latest version of CDM document Sampling and surveys for CDM project activities and programme of activities. Alternatively, the project participant may follow the simple random sampling approach and the minimum sample size should be determined as per the following guidelines:

- Project target population < 300: Minimum sample size 30
- Project target population 300 – 1000: Minimum sample size 10% of group size
- Project target population > 1000: Minimum sample size 100

This simplified approach was be used for determining minimum sample size for parameters listed in the section below in which case it was 95/10 precision.

#### Random Sampling for current monitoring period: -

| Sr. no | Parameter          | Description of Parameter                                                             | No of Samples (95/10) |
|--------|--------------------|--------------------------------------------------------------------------------------|-----------------------|
| 1      | $N_{y,i,j}$        | Number of project devices of type i and batch j operating during year y              | 108                   |
| 2      | $\mu_{y,i,j}$      | Adjustment to account for any continued use of pre-project devices during the year y | 25                    |
| 3      | $\eta_{new,y,i,j}$ | Efficiency of the device of each type i and batch j implemented as part of the       | 33                    |

ER Calculation for the current Project activity is given below:

$$B_{old,adjusted} = B_{old} \times (1 - \mu_y)$$

$$B_{old,adjusted} = (5.748) \times (1 - 0.0647) \\ = 5.376 \text{ tons wood-eq/HH-yr}$$

$$B_{y,savings,i,j} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i,j}}\right)$$

$$B_{y,savings,i,j} = ((5.376) \times (1 - (20/35.38))) \\ = 2.34 \text{ tons wood-eq/HH-yr}$$

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{woodfuel} \times (EF_{wf,co2} + EF_{wf, non co2}) \times N_{y,i,j} \times 0.95$$

$$ER_y = 2.34 * 0.0156 * 0.89 * (112 + 26.23) * 20478 * 0.95$$

$$= 12,978 \text{ tCO}_2\text{e}$$

| Year                                | Baseline emissions or removals (tCO <sub>2</sub> e) | Project emissions or removals (tCO <sub>2</sub> e) | Leakage emissions (tCO <sub>2</sub> e) | Net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|-------------------------------------|-----------------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------------------------------------------------|
| 2022(09-May-2022 to 31-August-2022) | 12,978                                              | -                                                  | -                                      | 12,978                                                       |
| Total                               | 12,978                                              | -                                                  | -                                      | 12,978                                                       |

| <u>Ex-ante emissions reductions/removals</u> | <u>Achieved emissions reductions/removals</u> | <u>Percent difference</u> | <u>Justification for the difference</u>                                                                                                                                                                                                                                                               |
|----------------------------------------------|-----------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 857,98 (115 days)                            | 12,978                                        | -83 %                     | Actual emission reductions achieved are lower than the value estimated in ex-ante calculation. This is mainly because the WBT test value derived from the sampling survey testing is lower than the projected value & very small year fraction of VCUs accounting for the project activity instances. |

# APPENDIX 1: METER DEVICES (CALIBRATION CERTIFICATES)

## Barometer

### Calibration details

|                  |                  |
|------------------|------------------|
| Calibration Date | 31- January 2022 |
| Due Date         | 31 January 2023  |
| Serial Number    | SGW-300H         |



**Keane Engineering Technologies Ltd**




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www.keaneengineering.co.ug

ISO/IEC17025:2017 Certificate No: CL/37 UNBS Recognition No: UNBS/LRS/0024

## Calibration Certificate

Cert No: KET/LAB/01/CAL/266 Page: 1 of 2

**1.0. CUSTOMER**  
 Name: Centre for Integrated Research and Community Development Uganda  
 Address: Plot 1715 Bbira Kireka, Bujuku Road P.O Box 16340 Wandegaya, Uganda  
 Telephone: +256392004309  
 Email: info@circoodu.org.ug  
 Calibration Site: Keane Laboratory  
 Job Number: KET/SRVS/CAL/JB374

**2.0. UNIT UNDER TEST/ CALIBRATION**  
 Name: Barometer  
 Manufacturer: Casio  
 Model/Type: WR100M  
 Serial Number: SGW-300H  
 Internal ID: None  
 Location: Laboratory  
 Calibration Sticker No: K167  
 Date of Receipt: 28/01/2022  
 Calibration Date: 31/01/2022

**3.0. REFERENCE STANDARDS USED**  
 Standard Reference Barometer Altimeter, with calibration certificate number: TNC/1422/11-20

**4.0. AMBIENT CONDITIONS**  
 Temperature: 20.0±1 °C Humidity: 51±1 %

**5.0. CALIBRATION PROCEDURE**  
 The Barometer has been calibrated as per procedure document No: DKD-R6-1.

**6.0. TRACEABILITY**  
 This Equipment has been calibrated using standard references traceable through TESTNCA Laboratory

**7.0. VALIDITY**  
 This certificate is valid until 31/01/2023 when the device will be due for re-calibration as per Customer request

Calibrated by: David Ssegulani Date: 31/01/2022  
 Prepared by: Wilfred Masaba Date: 02/02/2022  
 Checked & Authorized by:  Date: 03/02/2022  
 Technical Director

Calibration certificate without signature and official stamp is not valid. This certificate has been issued without any alteration and may not be reproduced except with the full approval of Technical Director

Cert No: KET/LAB/01/CAL/266 Page: 2 of 2

### 8.0 MEASUREMENT RESULTS

| Average Standard Reading (hPa) | Average Unit Under Test Reading (hPa) | Deviation (hPa) |
|--------------------------------|---------------------------------------|-----------------|
| 875.5                          | 877                                   | 1.5             |
| 839.7                          | 841                                   | 1.3             |
| 770.6                          | 772                                   | 1.4             |

**9.0. COMMENTS**

- The results in this certificate are correct at the time of Calibration. Subsequently the accuracy will depend on such factors as the care exercised in handling, Use of the Equipment and Frequency of use.
- The equipment has been calibrated and is within the Maximum permissible as specified by the manufacturer (±3% FSD).
- The Uncertainty of measurement is ± 4.5 hPa. This reported Uncertainty is stated as the expanded uncertainty of measurement multiplied by the coverage factor K=2, providing a confidence level of approximately 95%
- This Certificate relates to only this Equipment with Serial No.: SGW-300H

-----END OF CALIBRATION CERTIFICATE-----

Keane Engineering Technologies Ltd, Plot 432 Odur Lane along Kiwatule Road, Ntinda next to Eliana Hotel  
 P.O Box 34877, Kampala- Uganda, Tel: +256-200977778  
 Email: info@keaneengineering.co.ug Web: www.keaneengineering.co.ug

## Thermocouple

### Calibration details

|                  |                 |
|------------------|-----------------|
| Calibration Date | 31 January 2022 |
| Due Date         | 31 January 2023 |
| Serial Number    | 0490-258500     |



**Keane Engineering Technologies Ltd**




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www.keaneengineering.co.ug

ISO/IEC17025:2017 Certificate No: CL/37 UNBS Recognition No: UNBS/LRS/0024

### Calibration Certificate

Cert No: KET/LAB/06/CAL/164 Page...1...of...2...Pages

**1.0. CUSTOMER**  
 Name: Centre for Integrated Research and Community Development Uganda  
 Address: Plot 1715 Bbira Kireka, Bujjiko Road P.O Box 16340 Wandegaya, Uganda  
 Telephone: +256392004309  
 Email: info@cirocodu.org.ug  
 Calibration Site: Keane Laboratory  
 Job Number: KET/SRVS/CAL/JB374

**2.0. UNIT UNDER TEST/ CALIBRATION**  
 Name: Ethernet Multimeter  
 Manufacturer: Keithley  
 Model/Type: 2701  
 Serial Number: 0490-258500  
 Internal ID: None  
 Location: Laboratory  
 Calibration Sticker No: K065  
 Date of Receipt: 28/01/2022  
 Calibration Date: 31/01/2022

**3.0. REFERENCE STANDARDS USED**  
 a) Standard Reference Thermometer type SDL 200 with Calibration Certificate No: BS/MET/4/3/125/246  
 b) Standard Reference Liquid Bath type 6001091 with Calibration Certificate No: LSL/03/41/353

**4.0. AMBIENT CONDITIONS**  
 Temperature: 20±1 °C Humidity: 50±1 %

**5.0. CALIBRATION PROCEDURE**  
 The Equipment has been calibrated as per procedure document No: EURAMET Calibration Guide 8 Version 3

**6.0. TRACEABILITY**  
 This Equipment has been calibrated using standard references traceable through Kenya Bureau of Standards and LABCRS Services

**7.0. VALIDITY**  
 This certificate is valid until 31/01/2023 when the device will be due for re-calibration as per Customer request

Calibrated by: David Ssegulani Date: 31/01/2022  
 Prepared by: Wilfred Masaba Date: 02/02/2022  
 Checked & Authorized by:  Date: 03/02/2022

03 FEB 2022

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Page...2...of...2...pages

Cert No: KET/LAB/06/CAL/164

### 8.0 MEASUREMENT RESULTS

**Channel 2**

| Test Points in °C | Average Standard Reading in °C | Average Unit Under Test Reading in °C | Correction in °C | Uncertainty in (±) °C |
|-------------------|--------------------------------|---------------------------------------|------------------|-----------------------|
| 30                | 30.0                           | 29.69                                 | -0.31            | 0.70                  |
| 60                | 60.0                           | 60.10                                 | 0.10             | 0.61                  |
| 90                | 90.0                           | 90.88                                 | 0.88             | 1.18                  |

**Channel 3**

| Test Points in °C | Average Standard Reading in °C | Average Unit Under Test Reading in °C | Correction in °C | Uncertainty in (±) °C |
|-------------------|--------------------------------|---------------------------------------|------------------|-----------------------|
| 30                | 30.0                           | 29.55                                 | -0.45            | 0.79                  |
| 60                | 60.0                           | 59.92                                 | -0.08            | 0.61                  |
| 90                | 90.0                           | 90.75                                 | 0.75             | 1.05                  |

**Channel 4**

| Test Points in °C | Average Standard Reading in °C | Average Unit Under Test Reading in °C | Correction in °C | Uncertainty in (±) °C |
|-------------------|--------------------------------|---------------------------------------|------------------|-----------------------|
| 30                | 30.0                           | 29.10                                 | -0.90            | 1.20                  |
| 60                | 60.0                           | 59.75                                 | -0.25            | 0.67                  |
| 90                | 90.0                           | 90.10                                 | 0.10             | 0.61                  |

**9.0. COMMENTS**

- The results in this certificate are correct at the time of Calibration and the results apply to this equipment as received. Subsequently the accuracy will depend on such factors as the care exercised in handling, Use of the Equipment and Frequency of use.
- The Uncertainty of the calibration results for the measurement done is as stated in the table above. The reported expanded uncertainty is stated as expanded uncertainty of measurement multiplied by coverage factor K=2, providing a confidence level of approximately 95%.
- This Certificate relates to only this Equipment with Serial No.: 0490-258500

-----END OF CALIBRATION CERTIFICATE-----

Keane Engineering Technologies Ltd, Plot 432 Odur Lane along Kiwatule Road, Ntinda next to Eliana Hotel  
P.O. Box 34877, Kampala- Uganda, Tel: +256-200977778  
Email: info@keaneengineering.co.ug, Web: www.keaneengineering.co.ug



## Weighing Scale

### Calibration details

|                  |                 |
|------------------|-----------------|
| Calibration Date | 31 January 2022 |
| Due Date         | 31 January 2023 |
| Serial Number    | 1310049         |



**Keane Engineering Technologies Ltd**




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www.keaneengineering.co.ug

ISO/IEC17025:2017 Certificate No: CL/37 UNBS Recognition No: UNBS/LRS/0024

### Calibration Certificate

Page...1...of...2...Pages

**Cert No: KET/LAB/03/CAL/476**

**1.0. CUSTOMER**

Name: Centre for Integrated Research and Community Development Uganda  
Address: Plot 1715 Bbiri Kireka, Bujiko Road P.O Box 16340 Wandegaya, Uganda  
Telephone: +256392004309  
Email: info@circodu.org.ug  
Calibration Site: Keane Laboratory  
Job Number: KETSRVS/CAL/JB374

**2.0. UNIT UNDER TEST/CALIBRATION**

Name: Weighing Scale  
Manufacturer: None  
Model/Type: BLB16  
Serial Number: 1310049  
Internal ID: CIR/WBTS/01  
Maximum Capacity: 16Kg  
Resolution: 0.1Kg  
Location: Laboratory  
Sticker No: K377  
Date of Reception: 28/01/2022  
Date of Calibration: 31/01/2022

**3.0. REFERENCE STANDARDS USED**

Certified Standard Reference Weights, Class M1, with Calibration Certificate No: SCW-0219-046

**4.0. AMBIENT CONDITIONS**

Temperature: 20.8°C Humidity: 44.5%

**5.0. CALIBRATION PROCEDURE**

The Weighing Scale has been calibrated as per procedure document No: EURAMET Calibration Guide 18 Version 4.0

**6.0. TRACEABILITY**

This Weighing Scale has been calibrated against Standards traceable through MARUTI

**7.0. VALIDITY**

This certificate is valid until 31/01/2023 when the device will be due for re-calibration as per Customer request

Calibrated by: Hadadi Sseveri Date: 31/01/2022

Prepared by: Wilfred Masaba Date: 02/02/2022

Checked & Authorized by:  Date: 03/02/2022

03 FEB 2022

Calibration certificate without signature and official stamp is not valid. This certificate has been issued without any alteration and may not be reproduced except with the full approval of Technical Director

Page...2...of...2...pages

**Cert No: KET/LAB/03/CAL/476**

### 8.0 PREADJUSTMENT RESULTS

| Ref weight (g) | No | Standard Reading (g) | Scale Reading (g) | Deviation (g) |
|----------------|----|----------------------|-------------------|---------------|
| 500            | 1  | 500                  | 499.6             | -0.4          |
| 1000           | 2  | 1000                 | 999.8             | -0.2          |
| 2000           | 3  | 2000                 | 1999.8            | -0.2          |
| 5000           | 4  | 5000                 | 4999.7            | -0.3          |
| 10000          | 5  | 10000                | 9999.7            | -0.3          |
| 15000          | 6  | 15000                | 14999.4           | -0.6          |

**9.0 CALIBRATION RESULT**

**9.1 Linearity Test**

| Standard Reading (g) | 500.0 | 1000   | 2000   | 5000   | 10000   | 15000   |
|----------------------|-------|--------|--------|--------|---------|---------|
| Scale Reading (g)    | 500.0 | 1000.2 | 2000.2 | 4999.9 | 10000.1 | 15000.1 |
| Deviation (g)        | 0.0   | 0.2    | 0.2    | -0.1   | 0.1     | 0.1     |

**9.2 Repeatability Test**

| Test Weight (g) | No. of Tests | Average Scale Reading (g) | Deviation (g) | Standard Deviation (g) |
|-----------------|--------------|---------------------------|---------------|------------------------|
| 5000            | 10           | 4999.9                    | -0.1          | 0.0                    |

**9.3 Eccentricity Test (Standard Test Weight Used is 5000g)**

| Position          | Center | Left Rear | Right Rear | Right Front | Left Front | Centre |
|-------------------|--------|-----------|------------|-------------|------------|--------|
| Scale Reading (g) | 4999.8 | 4999.8    | 4999.8     | 4999.8      | 4999.8     | 4999.8 |
| Deviation (g)     | 0.0    | 0.0       | 0.0        | 0.0         | 0.0        | 0.0    |

**10.0 COMMENTS**

- The results in this certificate are correct at the time of Calibration and the results apply to this equipment as received. Subsequently the accuracy will depend on such factors as the care exercised in handling, Use of the Equipment and Frequency of use.
- The Uncertainty of the calibration results for the measurement done is  $\pm 0.02g$ . The reported expanded uncertainty is stated as expanded uncertainty of measurement multiplied by coverage factor  $K=2$ , providing a confidence level of approximately 95%.
- This Certificate relates to only this Equipment with Internal I.D No.: CIR/WBTS/01

-----END OF CALIBRATION CERTIFICATE-----

Keane Engineering Technologies Ltd, Plot 432 Odur Lane along Kiwatule Road, Ntinda next to Eliana Hotel  
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Email: info@keaneengineering.co.ug, Web: www.keaneengineering.co.ug

## APPENDIX 2: NO DOUBLE COUNTING (DECLARATION BY PROJECT DEVELOPER)



January 30, 2023

**Ref. Declaration for No Double Counting and ODA in the project VCS 2789**

Dear Sir/Madam

With reference to the above subject, UpEnergy Group confirms that the project VCS 2789 (UpEnergy - Social and Climate Impact Programme- Uganda-1)

- Is not registered with any other carbon standard,
- All the distributed ICS are unique to this project with appropriate serial numbers,

UpEnergy Group also confirms in the project VCS 2789 (UpEnergy - Social and Climate Impact Programme- Uganda-1) at any stage of project design or implementation

- No ODA has been used to support the development or implementation of the project, or that no entity is providing ODA to the host country and thus isn't liable in future to benefit directly or indirectly from the carbon credits generated from the above-mentioned project.

Sincerely,



Name: Anantha Karthik Rajagopalan  
Title: Director of Carbon Programme  
Company Name: UpEnergy Group

Named among Fast Company's 2014 Top 10 Most Innovative Companies in Africa

info@upenergygroup.com | Uganda • Mauritius • USA | <http://upenergygroup.com>



## APPENDIX 3: FNRB ASSESSMENT

PP has calculated the fNRB as 0.89 as per the guidelines outlined in CDM tool 30 v4 with the latest available data from the reliable data sources such as UN / FAO / IPCC etc as summarized below,

| # | Parameters                                                                                                                                                 | Obtained Value        | Source of Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Total consumption of woody biomass<br>( $H = HW \times N + CE + NE$ )                                                                                      | 13,35,45,800 (tonnes) | The total consumption of woody biomass (H) is calculated using the following equation, accounting for all consumption within the applicable area (not only wood fuel but also timber and industrial consumption):<br>Source:<br><u>UBOS, 2019. Statistical Abstract, 2019. Uganda Bureau of Statistics, Uganda. <a href="https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf">https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf</a></u>                                                                                                                                                                                                                            |
| 2 | Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period<br>MAI <sub>forest,natural forest</sub> | 0.50 (tonnes/ha/yr)   | Mean Annual Increment (MAI) of woody biomass growth<br>Woody biomass values for forest areas are sourced from IPCC report i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories<br><br>Source:<br><u>IPCC 2019. Chapter 4, Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). Published: IPCC, Switzerland. <a href="https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf">https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf</a></u> |
| 3 | Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period<br>MAI <sub>forest,wooded land</sub>    | 0.90 (tonnes/ha/yr)   | Mean Annual Increment (MAI) of woody biomass growth<br>Woody biomass values for forest areas are sourced from IPCC report i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories<br><br>Source:<br><u>IPCC 2019. Chapter 4, Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize S., Osako, A.,</u>                                                                                                                                                                                                                                                                                           |

| # | Parameters                                                                                                                                                                                   | Obtained Value          | Source of Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                                                                                              |                         | <u>Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds).</u><br>Published: IPCC, Switzerland. <a href="https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf">https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4 | Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period<br>MAI <sub>forest, plantations</sub>                                     | 7.95<br>(tonnes/h a/yr) | Mean Annual Increment (MAI) of woody biomass growth<br>Woody biomass values for forest areas are sourced from IPCC report i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories<br><br>Source:<br>IPCC 2019. Chapter 4, Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). Published: IPCC, Switzerland. <a href="https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf">https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf</a> |
| 5 | Mean Annual Increment of woody biomass growth per hectare in sub-category i of forest areas in the relevant period<br>MAI <sub>other, bushland+grassland</sub>                               | 0.90<br>(tonnes/h a/yr) | Mean Annual Increment (MAI) of woody biomass growth<br>Woody biomass values for forest areas are sourced from IPCC report i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories<br><br>Source:<br>IPCC 2019. Chapter 4, Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A., Baasansuren, J., Fukuda, M., Ngarize S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). Published: IPCC, Switzerland. <a href="https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf">https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch04_Forest%20Land.pdf</a> |
| 6 | Mean Annual Increment of woody biomass growth per hectare in sub-category i of other land areas in the relevant period<br>(MAI <sub>other, i</sub> )<br>(MAI <sub>other, agriculture</sub> ) | 0.47<br>(tonnes/h a/yr) | Mean Annual Increment (MAI) of woody biomass growth<br>Woody biomass values for other areas are sourced from IPCC report i.e. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories<br><br>Source:<br>IPCC 2019. Chapter 5, Cropland. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Calvo Buendia, E., Tanabe, K., Kranjc, A.,                                                                                                                                                                                                                                                                                                                                                   |

| # | Parameters                                                                                                                                                                                    | Obtained Value                                                  | Source of Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | land,                                                                                                                                                                                         |                                                                 | <u>Baasansuren, J., Fukuda, M., Ngarize S., Osako, A., Pyrozhenko, Y., Shermanau, P. and Federici, S. (eds). Published: IPCC, Switzerland. <a href="https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch05_Cropland.pdf">https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch05_Cropland.pdf</a></u>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7 | Extent of forest and other land area in sub-category i in the relevant period (ha)                                                                                                            | $F_{forest,i} = 89,44,915$ ha<br>$F_{other,i} = 1,05,30,8$ 20ha | <p>The land area details of forests and other lands with vegetation are sourced from FAO from Food &amp; Agriculture Organization of United Nations for the latest available vintage 2020</p> <p>Source</p> <p><u>FAO, 2020. Global Forest Resources Assessment 2020 Report, Uganda. Food and Agriculture Organization of the United Nations. Rome, 2020. <a href="http://www.fao.org/3/cb0084en/cb0084en.pdf">http://www.fao.org/3/cb0084en/cb0084en.pdf</a></u></p> <p>Source:</p> <p><u>UBOS, 2019. Statistical Abstract, 2019. Uganda Bureau of Statistics, Uganda. <a href="https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf">https://www.ubos.org/wp-content/uploads/publications/01_20202019_Statistical_Abstract_Final.pdf</a></u></p> |
| 8 | Extent of non-accessible area within forest and other land area in sub-category i in the relevant period (ha)                                                                                 | $P_{forest,i} = 7,31,000$ ha<br>$P_{other,i} = 0$ ha            | <p>- The information regarding protected forests and forests with long-term management plan are sourced from Global Forest Resources Assessment, 2020, FAO, Uganda</p> <p>Source:</p> <p><u>FAO 2015b. Global Forest Resources Assessment 2015. Country Report, Uganda. Food and Agriculture Organization of the United Nations, Rome, 2014. <a href="http://www.fao.org/3/a-az362e.pdf">http://www.fao.org/3/a-az362e.pdf</a></u></p>                                                                                                                                                                                                                                                                                                                                                        |
| 9 | Comparison of the calculated value of fNRB against the values for fNRB reported in relevant scientific literature (including the minimum requirements as per para. 6-a-i of the TOOL30 ver.4) | -                                                               | <p>The mean fNRB value from the quoted literature in CDM tool 30 works out to 61.3% vis-à-vis 89% based on the independent study conducted by a Forestry Expert on behalf of the PP as per CDM tool 30. This difference in the fNRB value can be attributed to the following reasons</p> <ul style="list-style-type: none"> <li>➤ The relevant scientific literature quoted as per CDM tool 30 (i) Bailis, R.; Drigo, R.; Ghilardi, A. &amp; Masera, O. (2015) is dated 2015 and the data points used in this research work is further older as 2009.</li> <li>➤ As a matter of fact, the fuelwood used for total energy supply at Uganda is 48,470 k cubic metre in 2009, which significantly increased to an extent 97% to 95,490 k</li> </ul>                                              |

| # | Parameters | Obtained Value | Source of Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |            |                | <p>cubic metre during 2020<br/>(Source : “Energy Statistics Database” from United Nation’s Database<br/><a href="https://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3AFW">https://data.un.org/Data.aspx?d=EDATA&amp;f=cmID%3AFW</a>)</p> <p>➤ On the other hand, total forest area in Uganda sums to be 2791.673 k hectare in 2009 and a sharp decline is noticed in recent years. In 2021 Uganda has only 2296.6433 k hectare of total forest area, this indicates ~15% of the total forest area is deforested as compared with 2009 scenario. (Source: “Food &amp; Agriculture Organization of United Nations”<br/><a href="https://www.fao.org/faostat/en/#data/RL">https://www.fao.org/faostat/en/#data/RL</a>)</p> <p>There are different ways to calculate fNRB including geo-spatial approach, but PP appointed an independent 3<sup>rd</sup> party expert to carry out extensive analysis to calculate the fNRB as per the guidelines outlined in CDM tool 30 v4 with the latest available data from the reliable data sources such as UN / FAO / IPCC etc.</p> |